Periodic Orbits and Dynamical Structures in the Earth-Moon Circular Restricted Three-Body Problem

MAE 541 Final Project

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Authors' Contributions

Section 2, Section 3, and Section 5 (and related Appendices) are the work of Claudio Vestini. Section 4 and Section 8 are the work of Jannik Gräbner. Section 6 and Section 7 (and related Appendix) are the work of Jack Sydney Schwartz (Section 7 is co-authored by Jannik Gräbner). The Introduction and Conclusion sections, Section 1 and Section 9, are the joint work of all three authors.

1 Introduction

Growing international and commercial interest in the cislunar (Earth–Moon) domain is driven by the goal of establishing a sustained human and robotic presence on and near the Moon. As a result, the number of planned missions in this region is increasing rapidly, with some projections estimating that nearly 50 missions to the cislunar environment will be launched by 2030, including efforts to support human exploration infrastructure, resource utilisation, and commercial activities [Joh22].

The cislunar environment is governed by complex dynamics, as objects move under the simultaneous gravitational influence of both the Earth and the Moon. In addition, they are affected by further perturbations, such as the gravitational pull of other bodies, including the Sun and Jupiter, as well as solar radiation pressure, drag, and other higher-order effects. The growing number of missions in this region requires a thorough understanding of its dynamical structure, which can be developed through mathematical analysis of the underlying system.

A common astrodynamics model used in preliminary mission design is the Circular Restricted Three-Body Problem (CR3BP). The CR3BP is a Hamiltonian system that describes the motion of a small body under the gravitational influence of two large bodies in a rotating frame of reference. Although it is a simplified model, it still captures many of the essential dynamical features of three-body systems.

Periodic orbits and dynamical structures of the CR3BP are particularly relevant to space mission design. In the Earth–Moon system, periodic orbits often serve as mission targets, and properties such as their stability and spatial location are often exploited to provide favourable conditions for the orbiting spacecraft. The stable and unstable manifolds of these orbits are often used in low-thrust trajectory design to approach or depart from the orbits in a fuel-efficient manner. The study of periodic orbits also supports the development of efficient long-term propagation methods for cislunar objects, for example, through normal form theory.

We study the CR3BP using several tools from dynamical systems theory. The outline of this paper is as follows. We begin by providing a brief historical overview and introducing the system dynamics in Section 2. In Section 3, we derive the equilibrium points, discuss their stability, and describe the associated Hill's regions. Section 4 follows by introducing periodic orbits in the CR3BP and outlining their computation using differential correction. Section 5 focuses on the invariant manifolds of these orbits, with particular emphasis on the numerics employed in their tracing. In Section 6, we visualise a large number of planar periodic orbits from different families, which appear as fixed points on a Poincaré section. The subsequent Section 7 employs another Poincaré map, now restricted to fixed-energy values, to illustrate the role of planar resonant orbits in structuring the surrounding dynamics. In Section 8, we use normal forms around periodic orbits to derive simplified representations of the local dynamics. Finally, Section 9 concludes the analysis.

2 The Circular Restricted Three-Body Problem

Our project studies the dynamics of the CR3BP. A simplification of the more general Three-Body Problem (3BP), this continuous-time, nonintegrable dynamical system is one of the most celebrated and extensively studied in the history of both mathematics and physics. It is, in fact, through the analysis of this very system that French mathematician Henri Poincaré gave birth to the fields of dynamical systems and chaos theory.

2.1 A Brief Historical Summary

Although problems in astrodynamics have forever inspired humankind and his discoveries, the first recognisable mathematical formalisation of the general 3BP appeared in the XVII century *Principia Mathematica* by Isaac Newton [New87]. After successfully inventing the mathematical framework to generalise previous findings on the Two-Body problem (i.e. the Keplerian orbits and Kepler's laws), Newton elegantly derived the dynamics of the Earth–Moon–Sun system, and attempted to find admitted closed-form solutions, without success. In conversations with his contemporary Dr Edmond Halley, who was pressing Newton to finish his work on 3BP theory, he admitted that studying this system "made [his] head ache, and kept [him] awake so often, that [he] would think of it no more" [Bre55].

Newton's work was picked up in the 1760's by Leonhard Euler, who took the first major steps toward simplifying the problem to the modern CR3BP form. He did so by introducing a rotating coordinate system that moved along with the two larger bodies. Euler was also the first to search for conditions under which a third body could remain fixed relative to the other two, discovering the first three *collinear* equilibrium points [Eul67].

A comprehensive mathematical formalisation of the restricted assumption came from Joseph-Louis Lagrange in his 1772 *Essai sur le problème des trois corps* [Lag72]. Lagrange also discovered two additional *triangular* equilibrium points. As a result, today all five equilibrium points of the CR3BP are universally referred to as the Lagrange points.

For decades to follow, mathematicians searched for integrals of motion that would allow the differential equations to be solved in closed-form. In 1836, Carl Gustav Jacob Jacobi formulated the existence of the first integral of motion of the CR3BP, the *Jacobi constant*, which remains the only known conserved quantity [Jac36]. Using his constant, Jacobi also introduced *zero-velocity surfaces*, defining the set of reachable states for the third body.

In the mid-to-late XIX century George William Hill provided a formal analysis of the zero-velocity surfaces, earning himself the naming of their mathematical complements as *Hill's regions* [Hil78]. By analysing the topological closure of these unreachable regions, Hill proved that the distance between the Earth and the Moon is strictly bounded, establishing that the Moon cannot escape Earth's gravitational influence to become a heliocentric body.

The late XIX century marked a profound shift in how the CR3BP was understood. Motivated by an 1887 prize offered by King Oscar II of Sweden for a general mathematical solution to the n-body problem, Henri Poincaré submitted a manuscript focusing heavily on the CR3BP. In this monumental work, Poincaré proved mathematically that the system is nonintegrable, thus establishing that a closed-form algebraic solution is impossible [Poi90]. Instead, Poincaré shifted the focus of the analysis towards topological and qualitative dynamics. In so doing, he first discovered *homoclinic tangles*, giving rise to the modern field of chaos theory. In addition, Poincaré was also the first to theorise the existence of an infinite family of periodic orbits admitted by the CR3BP [Poi93]. These were later traced out with the advent of digital computation and numerical methods.

Indeed, most recent contributions to the CR3BP have been enabled by the application of modern computational solvers to Poincaré's theory. At the dawn of the XXI century, researchers were able to identify and map the stable and unstable *invariant manifolds* that branch from periodic orbits [KLMR00], demonstrating their utility for optimal mission design through an interplanetary transport network of these manifolds. The 3BP is increasingly receiving attention even outside the academic sphere: in 2024, the TV series *3 Body Problem*, an English-language adaptation of a book by Chinese author Liu Cixin, was

released and quickly became viral on the streaming site Netflix. To this day, the CR3BP remains an active frontier of scientific research. This simplified system is often used as a preliminary framework for orbital mission design, serving as a bridge between analytical theory and the modern demands of space exploration.

2.2 Dynamics and Notation

The CR3BP describes the motion of a body of negligible mass m under the gravitational attraction of two larger bodies. The larger bodies are referred to as the *primary*, with normalised mass m_1, and the *secondary*, with normalised mass m_2, with $m_1 > m_2$. A system of natural units is used to normalise position, time, and velocity variables: the distance unit (DU) is the distance between the primary and secondary, the time unit (TU) is the orbital period of the primary and secondary per unit radian, and the mass unit (MU) is the total mass of the system (see Appendix A for detail on normalisation of units).

The only parameter of the system is the *mass ratio*, defined as

$$\mu = \frac{m_2}{m_1 + m_2}. \tag{2.1}$$

All constants for the Earth–Moon system are shown in Table 1. The dynamics are described in a *synodic* (rotating) frame of reference (x, y, z), shown in Fig. 1, with the primary and secondary located on the x–axis at x values of $-\mu$ and $1-\mu$, respectively.

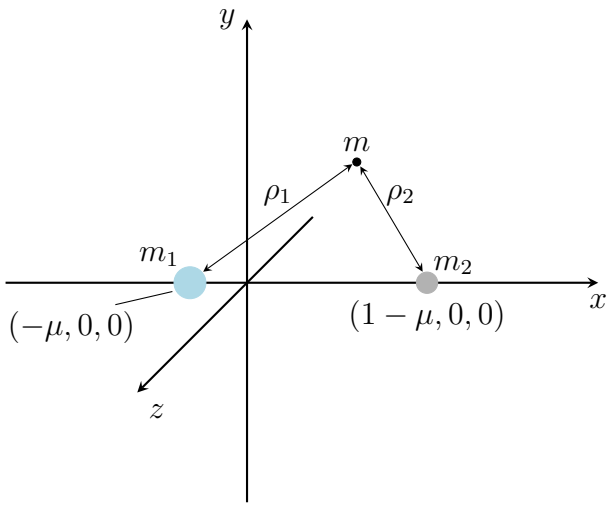


Figure 1: Illustration of the CR3BP in the synodic frame.

Table 1: Normalisation constants for the Earth–Moon CR3BP.

Quantity	Value
μ	1.215×10^{-2}
DU	3.844×10^{5} km
TU	2.361×10^{6} s
MU	6.0458×10^{24} kg

Denoting the position and velocity vectors as

$$\boldsymbol{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \qquad \boldsymbol{v} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix},$$

the CR3BP equations of motion can be written as the second-order nonlinear system

$$\ddot{\boldsymbol{r}} = \boldsymbol{g}(\boldsymbol{r}, \boldsymbol{v}) = \begin{pmatrix} 2v_y + x - (1-\mu)\dfrac{x+\mu}{\rho_1^3} - \mu\dfrac{x-1+\mu}{\rho_2^3} \\ -2v_x + y - (1-\mu)\dfrac{y}{\rho_1^3} - \mu\dfrac{y}{\rho_2^3} \\ -(1-\mu)\dfrac{z}{\rho_1^3} - \mu\dfrac{z}{\rho_2^3} \end{pmatrix}, \tag{2.2}$$

where

$$\rho_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \qquad \rho_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}$$

are the distances from the third body to the primary and secondary, respectively (see Fig. 1). A full derivation of the system dynamics in Eq. (2.2) from first principles, as well as a discussion of the system's symmetries, is provided in Appendix A.

The dynamics can be rewritten as a first-order autonomous (time-independent) system. We define the full state vector in position–velocity coordinates as

$$\boldsymbol{x} \coloneqq \begin{pmatrix} \boldsymbol{r} \\ \boldsymbol{v} \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \\ v_x \\ v_y \\ v_z \end{pmatrix}. \tag{2.3}$$

The system's symmetries described in Appendix A allow for a Planar reduction[1] to the CR3BP (PCR3BP). In the sections that consider the PCR3BP, we use the same notation but omit z and v_z from the state vector. The dynamics in Eq. (2.2) can be written compactly as

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) = \begin{pmatrix} \dot{\boldsymbol{r}} \\ \dot{\boldsymbol{v}} \end{pmatrix} = \begin{pmatrix} \boldsymbol{v} \\ \boldsymbol{g}(\boldsymbol{r}, \boldsymbol{v}) \end{pmatrix}. \tag{2.4}$$

For an initial condition $\boldsymbol{x}_0$, we denote the solution of

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}), \qquad \boldsymbol{x}(0) = \boldsymbol{x}_0,$$

as the flow map $\boldsymbol{\phi}$, written as

$$\boldsymbol{x}(t) = \boldsymbol{\phi}(t, \boldsymbol{x}_0),$$

which maps an initial state to the state reached after time t.

3 Equilibrium Points and Hill's Regions

3.1 Hamiltonian Reformulation

The equations of motion in Eq. (2.2) derived in Appendix A can be reformulated in terms of the gradients of scalar fields, or *potentials*. In particular, the inertial and centrifugal terms (the two conservative terms) in the inertial transport theorem (see Eq. (A.2)) are associated with a gravitational and centrifugal potential, respectively. Denoting these as U_g and U_c, respectively, these potentials are expressed as

$$\begin{aligned} U_g &= -\frac{1-\mu}{\rho_1} - \frac{\mu}{\rho_2}, \\ U_c &= -\frac{1}{2}(\boldsymbol{\omega} \times \boldsymbol{r}) \cdot (\boldsymbol{\omega} \times \boldsymbol{r}) = -\frac{1}{2}\left(x^2 + y^2\right). \end{aligned}$$

Combining the two yields the *effective potential*, $\tilde{U} \coloneqq U_g + U_c$. These potentials are visualised in Fig. 2 and serve as important tools for the analysis of the CR3BP. The potentials are symmetric with respect to the z coordinate (consistently with symmetry 1 in Appendix A).

[1] This reduction is consistent with symmetry 1 in Appendix A, since the plane $z = v_z = 0$ is invariant under the CR3BP dynamics Eq. (2.2); that is, an initial condition satisfying $z(0) = v_z(0) = 0$ remains planar for all time.

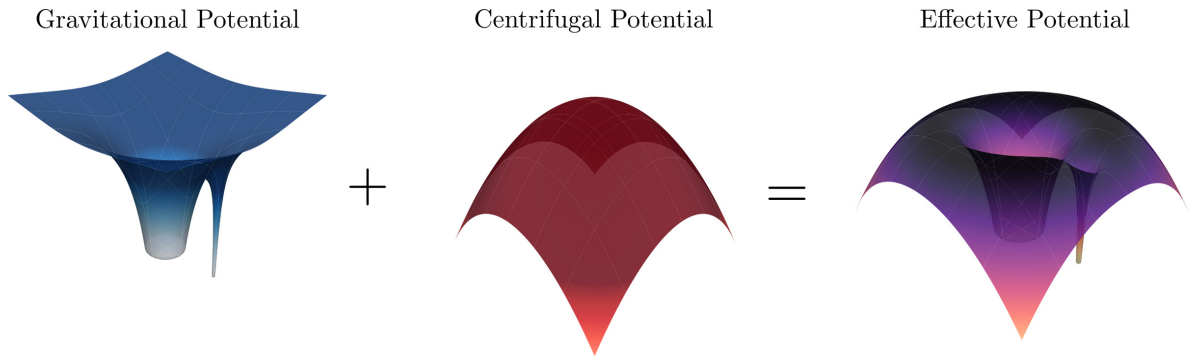


Figure 2: Illustration of scalar potentials as functions over the (x, y) plane.

Since the system is closed, the total energy of the smallest body m in the synodic frame is conserved. We identify the total energy as the sum of kinetic and potential energies. This quantity is the Hamiltonian of the system in Eq. (2.2), written as

$$H = \frac{1}{2}\|\boldsymbol{v}\|_2^2 + \tilde{U} = \frac{1}{2}\left(v_x^2 + v_y^2 + v_z^2\right) - \frac{1-\mu}{\rho_1} - \frac{\mu}{\rho_2} - \frac{1}{2}\left(x^2 + y^2\right). \tag{3.1}$$

It can be shown that the dynamics Eq. (2.2) can be rewritten in terms of the Hamiltonian as

$$\begin{aligned} \dot{\boldsymbol{r}} &= \nabla_{\boldsymbol{v}} H, \\ \dot{\boldsymbol{v}} &= \begin{pmatrix} 0 & 2 & 0 \\ -2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \nabla_{\boldsymbol{v}} H - \nabla_{\boldsymbol{r}} H, \end{aligned} \tag{3.2}$$

where the first term in the $\dot{\boldsymbol{v}}$ equation represents the Coriolis acceleration. We remark that this term can be eliminated by transforming to *canonical coordinates*; in these coordinates, the velocity $\boldsymbol{v}$ is replaced by canonical momentum $\boldsymbol{p}$, which, in the rotating frame, is given by $\boldsymbol{p} = (v_x - y, v_y + x, v_z)$. In $\boldsymbol{z} := (\boldsymbol{r}, \boldsymbol{p})$ coordinates, the system satisfies Hamilton's canonical equation

$$\dot{\boldsymbol{z}} = \boldsymbol{J}\nabla H(\boldsymbol{z}), \qquad \boldsymbol{J} = \begin{pmatrix} \boldsymbol{0}_{3\times 3} & \boldsymbol{I}_3 \\ -\boldsymbol{I}_3 & \boldsymbol{0}_{3\times 3} \end{pmatrix}, \tag{3.3}$$

where $\boldsymbol{J}$ is the canonical symplectic matrix, $\boldsymbol{I}_3$ is the 3×3 identity matrix, and $\boldsymbol{0}_{3\times 3}$ is the 3×3 matrix of zeros. The CR3BP system is therefore Hamiltonian.

3.2 Derivation of Lagrange Points

Armed with the reformulated system in Eq. (3.2), we derive the existence and coordinates of all equilibrium points. Following the standard approach in MAE 541, this can be done by imposing the condition $\dot{\boldsymbol{r}} = \dot{\boldsymbol{v}} = \boldsymbol{0}_3$, where $\boldsymbol{0}_3$ is the 3×1 vector of zeros. Applying this to the position equation in Eq. (3.2) yields $\nabla_{\boldsymbol{v}} H = 0$, simplifying the velocity equation to $-\nabla_{\boldsymbol{r}} H = 0$. Substituting the Hamiltonian definition Eq. (3.1), we obtain $\nabla_{\boldsymbol{r}} H = \nabla_{\boldsymbol{r}} \tilde{U} = 0$.

Thus, the condition for fixed points is solely dependent on the spatial gradient of the effective potential $\nabla_{\boldsymbol{r}} \tilde{U}$. Visually, the fixed points will correspond to "flat" locations of

the surface in the rightmost panel of Fig. 2. Mathematically, we can express this as

$$\begin{aligned} 0 &= x - \frac{1-\mu}{\rho_1^3}(x+\mu) - \frac{\mu}{\rho_2^3}(x-1+\mu), \\ 0 &= \left[1 - \frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3}\right] y, \\ 0 &= \left[-\frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3}\right] z. \end{aligned} \tag{3.4}$$

From the third equation in Eq. (3.4), we notice that the two terms within square brackets are always strictly negative; thus, this equation implies $z = 0$ for all fixed points (i.e. equilibrium points lie in the (x, y) plane (again consistently with symmetry 1 in Appendix A). The second equation in Eq. (3.4) is satisfied in the two following cases:

1. If $y = 0$;
2. If $y \neq 0$ and $[1 - (1-\mu)/\rho_1^3 - \mu/\rho_2^3] = 0$.

Case 1 defines the collinear equilibrium points $\mathcal{L}_1, \mathcal{L}_2$, and $\mathcal{L}_3$, whereas case 2 describes the triangular equilibrium points $\mathcal{L}_4$ and $\mathcal{L}_5$. Collinear points are located on the x–axis. By setting $y = 0$ (case 1) and $z = 0$, the first equation in Eq. (3.2) simplifies to:

$$x - \frac{(1-\mu)(x+\mu)}{|x+\mu|^3} - \frac{\mu(x-1+\mu)}{|x-1+\mu|^3} = 0.$$

Introducing the variable $u \coloneqq x - 1 + \mu$ leads to the identity $x + \mu = u + 1$. Consequently, the distances from m to primary and secondary simplify to $\rho_1 = |u+1|$ and $\rho_2 = |u|$, respectively. Substituting these into the first equilibrium equation in Eq. (3.4) yields:

$$u + 1 - \mu - \frac{(1-\mu)(u+1)}{|u+1|^3} - \frac{\mu u}{|u|^3} = 0.$$

By defining signum variables $s_0 \coloneqq \operatorname{sgn}(u)$ and $s_1 \coloneqq \operatorname{sgn}(u+1)$, this expression can be condensed to:

$$u + 1 - \mu - \frac{(1-\mu)s_1}{(u+1)^2} - \frac{\mu s_0}{u^2} = 0.$$

After algebraic manipulation of the above, we obtain the *Lagrange quintic equation* [Lag72]

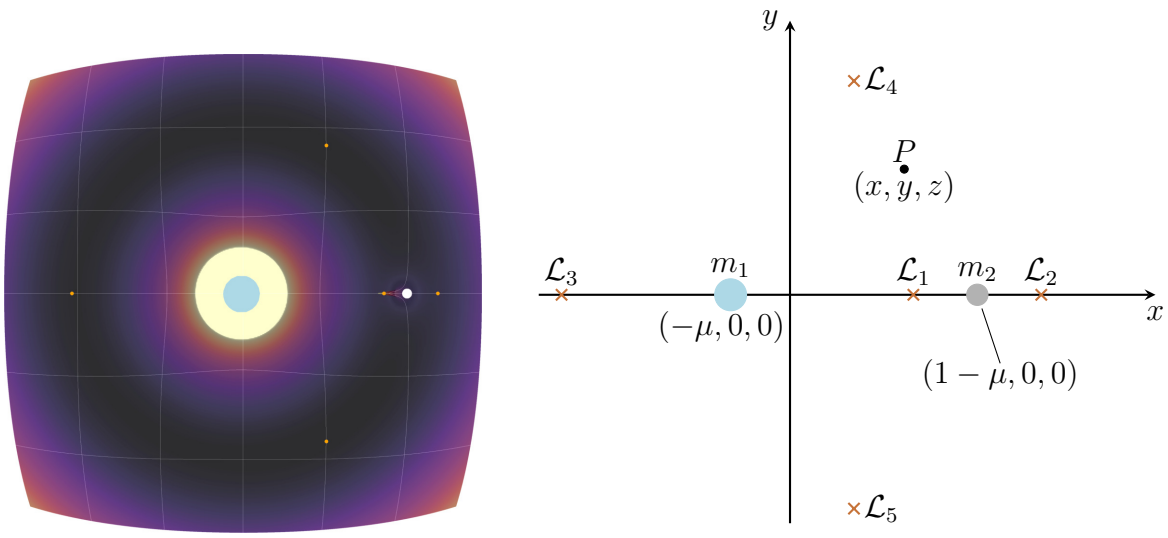
$$u^2(1 - s_1 + 3u + 3u^2 + u^3) = \mu[s_0 + 2s_0u + (1 + s_0 - s_1)u^2 + 2u^3 + u^4]. \tag{3.5}$$

The signs of s_0 and s_1 vary based on the position along the x–axis, identifying the following three distinct intervals:

$$\begin{aligned} &\mathcal{L}_3 : (s_0, s_1) = (-1, -1) \quad \text{for } x < -\mu, \\ &\mathcal{L}_1 : (s_0, s_1) = (-1, 1) \quad \text{for } -\mu < x < 1 - \mu, \\ &\mathcal{L}_2 : (s_0, s_1) = (1, 1) \quad \text{for } x > 1 - \mu. \end{aligned}$$

On each of these intervals, the Lagrange quintic equation Eq. (3.5) simplifies to:

$$\begin{aligned} &\mathcal{L}_1 : u^2(u^3 + 3u^2 + 3u) = \mu(u^4 + 2u^3 - u^2 - 2u - 1), \\ &\mathcal{L}_2 : u^2(u^3 + 3u^2 + 3u) = \mu(u^4 + 2u^3 + u^2 + 2u + 1), \\ &\mathcal{L}_3 : u^2(u^3 + 3u^2 + 3u + 2) = \mu(u^4 + 2u^3 + u^2 - 2u - 1). \end{aligned} \tag{3.6}$$



(a) Numerical solution. (b) Annotated plot.

Figure 3: Illustration of all five Lagrange points in the PCR3BP.

Each of the equations in Eq. (3.6) admits one real solution and two pairs of complex-conjugate solutions [Lag72]. Since these solutions correspond to the x–axis coordinate of each collinear equilibrium point, we retain the three real solutions, corresponding to the $\mathcal{L}_1, \mathcal{L}_2$, and $\mathcal{L}_3$ libration points. These real-valued solutions to Eq. (3.6) can be obtained in closed-form only through the adoption of elliptic functions. Instead, by means of numerical root-finding algorithms, we can compute approximate solutions. In the Earth–Moon system, the location of the collinear Lagrange points is computed as

$$\begin{aligned}
\mathcal{L}_1 &\approx (0.8369, 0, 0)\,, \\
\mathcal{L}_2 &\approx (1.1557, 0, 0)\,, \\
\mathcal{L}_3 &\approx (-1.0051, 0, 0)\,.
\end{aligned}$$

The remaining two libration points are found by analysing case 2, which gives the simultaneous system

$$\begin{aligned}
0 &= x - \frac{1-\mu}{\rho_1^3}(x+\mu) - \frac{\mu}{\rho_2^3}(x-1+\mu), \\
0 &= 1 - \frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3}.
\end{aligned}$$

By multiplying the second expression by $(x - 1 + \mu)$, subtracting it from the first, and simplifying, we obtain:

$$\frac{1-\mu}{\rho_1^3} = 1 - \mu,$$

which leads directly to $\rho_1 = 1$. Similarly, multiplying the second equation by $x + \mu$, subtracting it from the first one, and simplifying, yields

$$\frac{\mu}{\rho_2^3} = \mu,$$

which gives $\rho_2 = 1$. These two points therefore lie on the off-x–axis vertices of two unitary

equilateral triangles (hence their name), at coordinates

$$\mathcal{L}_4 = \left(\frac{1}{2} - \mu, \frac{\sqrt{3}}{2}, 0\right),$$
$$\mathcal{L}_5 = \left(\frac{1}{2} - \mu, -\frac{\sqrt{3}}{2}, 0\right).$$

It is interesting to note that, despite the triangular points having been derived (by Lagrange) 12 years after the collinear ones (which were derived by Euler, see Section 2.1), the former set is actually much simpler to obtain in practice (through mere algebraic reformulation). The five equilibrium points for the Earth–Moon PCR3BP in the synodic reference frame are illustrated in Fig. 3 (since all five points lie in the (x, y) plane, the z coordinate axis has been omitted from this illustration). The left panel shows a numerical solution overlaid onto the effective potential surface, whereas the right panel shows an annotated plot. The Lagrange points are plotted as yellow dots (left panel) and crosses (right panel). It is important to note that, since the locations of these points are fixed in the synodic frame, the Lagrange points orbit alongside the primary and secondary.

3.3 Stability

The collinear equilibrium points $\mathcal{L}_1, \mathcal{L}_2$, and $\mathcal{L}_3$ exhibit a *centre* $\times$ *centre* $\times$ *saddle* stability profile, rendering them unstable for all values of μ. The triangular equilibrium points $\mathcal{L}_4$ and $\mathcal{L}_5$ display a *centre* $\times$ *centre* $\times$ *centre* behaviour provided that $\mu < \mu_R$, where μ_R denotes the *Routh mass parameter*, which takes numerical value $\mu_R \approx 0.0385$. Due to page count limitations for this report, and in order not to interrupt or derail the discussion, the stability analysis has been embedded in Appendix B.

3.4 Jacobi Constant and Hill's Regions

In the astrodynamics literature, the total energy of the CR3BP is often referred to in terms of the system's first integral of motion, the *Jacobi integral* C, which can be written in terms of the Hamiltonian as

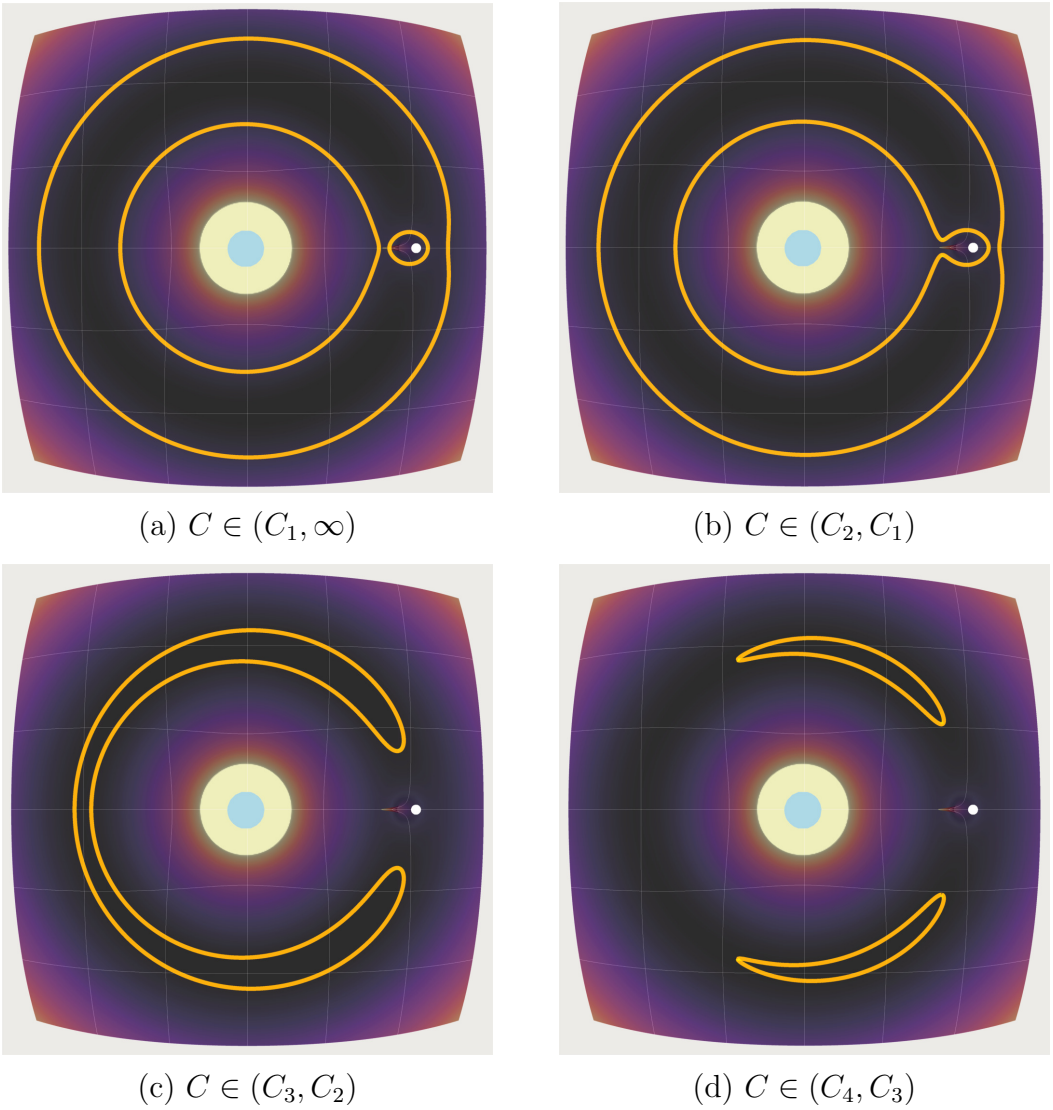
$$C = -2H. \tag{3.7}$$

The conservation law for C can be derived by integrating the quantity $\ddot{\boldsymbol{r}} \cdot \boldsymbol{v}$, and it results in topological bounds on the reachable states for body m [Jac36]. In particular, since the kinetic energy must be a nonnegative real value, body m is spatially constrained to *reachable regions*, defined by the condition $\tilde{U} < -C/2$. The boundaries of the reachable regions, defined by the condition of zero kinetic energy ($\tilde{U} = -C/2$), are known as *zero velocity surfaces*, and are defined on the full state space (x, y, z, v_x, v_y, v_z). On the other side of these boundaries, the set of unreachable states (including their boundary) is referred to as the *Hill's region*, defined on the spatial subspace (x, y, z) as:

$$\left\{(x, y, z) : \tilde{U}(x, y, z) \geq -\frac{1}{2}C\right\}.$$

The Hamiltonian and Jacobi integral values for each Lagrange point $\mathcal{L}_i$ ($i \in \{1, 2, 3, 4, 5\}$), are computed by substituting the fixed points' coordinates into Eq. (3.1) with $\boldsymbol{v} = \boldsymbol{0}_3$. They are summarised in Table 2, and the Hill's regions associated with each range are visualised in Fig. 4. These Hill's regions are computed numerically by considering "horizontal" cross sections of the effective potential surface $\tilde{U}$ at several "vertical" constant values of C.

Contours are then plotted as dark-yellow outlines in Fig. 4, and the Hill's regions lie on the "inside" of these outlines. These figures show top-down views of the potential surface, constraining the data to the (x, y) plane (with the same axes orientation used in previous figures). Thus, third body m cannot reach states inside the yellow outlines. The constant C values for each figure are chosen in such a way as to capture structurally different regimes, following from the threshold values in Table 2.



(a) $C \in (C_1, \infty)$ (b) $C \in (C_2, C_1)$

(c) $C \in (C_3, C_2)$ (d) $C \in (C_4, C_3)$

Figure 4: Hill's regions across Jacobi constant over the (x, y) plane.

Table 2: Hamiltonian and Jacobi integral values at the five equilibrium points for the Earth-Moon CR3BP.

Equilibrium Point	Coordinates (x, y, z)	H_i	C_i
$\mathcal{L}_1$	$(0.8369, 0, 0)$	-1.5941	3.1882
$\mathcal{L}_2$	$(1.1557, 0, 0)$	-1.5861	3.1722
$\mathcal{L}_3$	$(-1.0051, 0, 0)$	-1.5061	3.0122
$\mathcal{L}_4$	$(0.4878, 0.8660, 0)$	-1.4940	2.9880
$\mathcal{L}_5$	$(0.4878, -0.8660, 0)$	-1.4940	2.9880

From Fig. 4a we can see that, for $C > C_1$, m is constrained to a quasi-circular region around the primary m_1 (which is shown in light blue). As the effective potential value is increased (and thus the value of C decreased), a gateway to $\mathcal{L}_1$ begins to open. For $C_1 > C > C_2$, m gains access to the secondary m_2 (which is shown in white) via the $\mathcal{L}_1$ gateway, but cannot yet access $\mathcal{L}_2$, as shown in Fig. 4b. A second gateway to $\mathcal{L}_2$ opens for $C_2 > C > C_3$, and for this range of energy, m can escape the Earth–Moon system, as visible in Fig. 4c. Access to $\mathcal{L}_3$ is gained once $C_3 > C > C_4 = C_5$, at which point the only regions inaccessible to m are boomerang-shaped areas around the triangular points, as we can see from Fig. 4d. Finally, for energy values $C < C_4 = C_5$, m gains access to the entire domain (which we did not plot since the Hill's region coincides with the empty set).

4 Computing Periodic Orbits

4.1 Families of Periodic Orbits

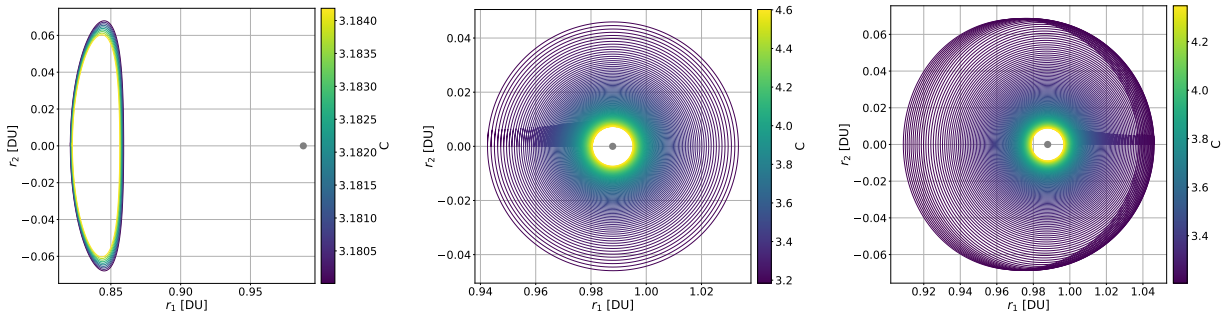
The CR3BP admits infinitely many periodic orbits [Poi93]. Many of these orbit families have been studied extensively and several have found practical applications in space mission design. In this work, we restrict our analysis to planar periodic orbits and illustrate how they can be computed numerically using differential correction. An initial condition $\boldsymbol{x}_0$ lies on a periodic orbit if there exists a time $T > 0$ such that

$$\boldsymbol{\phi}(T, \boldsymbol{x}_0) = \boldsymbol{x}_0.$$

The smallest positive value of T is called the period of the orbit.

Starting from an initial guess for the state $\boldsymbol{x}_0$, periodic orbits can be computed using a differential correction algorithm. Fig. 5 shows representative planar periodic orbits in the lunar region. These include Moon-centred orbit families, such as distant retrograde orbits (DROs) and distant prograde orbits (DPOs), as well as equilibrium-point-centred families, such as Lyapunov orbits.

Another important class of planar periodic orbits in the CR3BP is given by resonant orbits. A $p : q$ resonant orbit completes p revolutions around the Earth in the same time that the Moon completes q revolutions around the Earth, where p and q are integers. Depending on the direction of motion around the Earth, resonant orbits can be either prograde or retrograde. Examples of several resonant orbit families are shown in Fig. 6.



(a) Lyapunov (b) DRO (c) DPO

Figure 5: Examples of different families of orbits in the lunar regime, shown in the (x, y) plane of the CR3BP. Colours indicate the Jacobi constant for each orbit.

4.2 Differential Correction

Differential correction is a Newton-type method for refining an approximate trajectory until it satisfies a prescribed set of boundary conditions. Many trajectory-design problems can be formulated as finding a vector of free variables $\boldsymbol{X}$ such that

$$\boldsymbol{F}(\boldsymbol{X}) = \boldsymbol{0},$$

where $\boldsymbol{F}$ is a constraint vector measuring the violation of the desired boundary conditions. For example, in the simplest periodic-orbit problem, the free variables may be the initial condition $\boldsymbol{x}_0$ and the period T, and the constraint is

$$\boldsymbol{F}(\boldsymbol{x}_0, T) = \boldsymbol{\phi}(T, \boldsymbol{x}_0) - \boldsymbol{x}_0.$$

A periodic orbit is obtained when this residual vanishes.

Starting from an initial guess $\boldsymbol{X}^{(k)}$, differential correction linearises the constraint vector as

$$\boldsymbol{F}(\boldsymbol{X}^{(k)} + \delta\boldsymbol{X}) \approx \boldsymbol{F}(\boldsymbol{X}^{(k)}) + D\boldsymbol{F}(\boldsymbol{X}^{(k)})\delta\boldsymbol{X},$$

where $D\boldsymbol{F}(\boldsymbol{X}^{(k)})$ is the Jacobian of the constraints with respect to the free variables. The correction $\delta\boldsymbol{X}$ is chosen so that the linearised residual is zero:

$$D\boldsymbol{F}(\boldsymbol{X}^{(k)})\delta\boldsymbol{X} = -\boldsymbol{F}(\boldsymbol{X}^{(k)}). \tag{4.1}$$

The free variables are then updated according to

$$\boldsymbol{X}^{(k+1)} = \boldsymbol{X}^{(k)} + \delta\boldsymbol{X}. \tag{4.2}$$

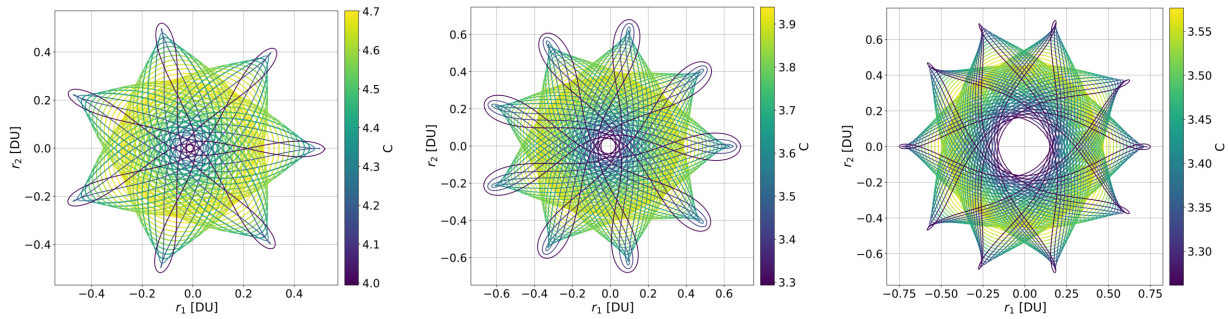
This process is repeated until the constraint residual satisfies

$$\left\| \boldsymbol{F}(\boldsymbol{X}^{(k)}) \right\| < \varepsilon,$$

for a prescribed tolerance ε.

In trajectory-design applications, the Jacobian $D\boldsymbol{F}$ is obtained from sensitivities of the flow map with respect to the free variables. In particular, the sensitivity of a state at time t with respect to the initial state is given by the state transition matrix (STM)

$$\boldsymbol{\Phi}(t, \boldsymbol{x}_0) = \frac{\partial \boldsymbol{\phi}(t, \boldsymbol{x}_0)}{\partial \boldsymbol{x}_0}.$$



(a) 7:1 Prograde (b) 9:2 Prograde (c) 10:3 Retrograde

Figure 6: Examples of different resonant orbit families, shown in the x, y plane of the CR3BP. Colours indicate the Jacobi constant for each orbit.

The STM is computed by integrating the variational equations along the current trajectory,

$$\dot{\boldsymbol{\Phi}}(t) = D\boldsymbol{f}(\boldsymbol{x}(t))\boldsymbol{\Phi}(t), \qquad \boldsymbol{\Phi}(0) = I. \tag{4.3}$$

For the PCR3BP, the matrix $D\boldsymbol{f}$ is given by

$$D\boldsymbol{f}(\boldsymbol{x}) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \tilde{U}_{xx} & \tilde{U}_{xy} & 0 & 2 \\ \tilde{U}_{xy} & \tilde{U}_{yy} & -2 & 0 \end{pmatrix}, \tag{4.4}$$

where the derivatives of the augmented potential are defined in Eq. (B.1) in the Appendix. Thus, each differential-correction iteration requires integrating both the equations of motion and the associated variational equations.

4.3 Computation of Planar Resonant Orbits

Since resonant orbits form one of the largest families of periodic orbits in the PCR3BP, we describe their computation using differential correction in more detail. The resonant orbits considered in this work are computed using `pydylan`, a Python interface to the Dynamically Leveraged Automated (N) Multibody Trajectory Optimisation software [BSJ+22]. For this purpose, we modified parts of the existing `pydylan` implementation related to the initialisation and differential correction of resonant orbits. Due to the symmetries of the PCR3BP, resonant orbits can be initialised at a crossing of the x–axis with $v_x = 0$. This crossing point is used as the initial state on the orbit and has the form

$$\boldsymbol{x}_0 = \begin{pmatrix} x_0 \\ 0 \\ 0 \\ v_{y,0} \end{pmatrix}.$$

Each family of resonant orbits exists over a range of x_0 values. To compute a specific member of such a family, x_0 is fixed, and the differential correction problem reduces to solving for the corresponding initial velocity $v_{y,0}$ and orbit period T.

To improve numerical stability, the periodicity condition is imposed using the symmetry of PCR3BP orbits. Rather than correcting the trajectory over the full period, the initial condition is propagated for a half-period $T/2$, and the trajectory is required to reach the x–axis orthogonally. Therefore, the free-variable vector in the differential correction algorithm is

$$\boldsymbol{X} = \begin{pmatrix} v_{y,0} \\ T/2 \end{pmatrix}.$$

Using a single-shooting transcription, the initial condition is propagated to

$$\boldsymbol{x}_f = \boldsymbol{\phi}(T/2, \boldsymbol{x}_0) = \begin{pmatrix} x(T/2) \\ y(T/2) \\ v_x(T/2) \\ v_y(T/2) \end{pmatrix}.$$

The half-period symmetry conditions are enforced through the differential-correction constraint vector

$$\boldsymbol{F}(\boldsymbol{X}) = \begin{pmatrix} y(T/2) \\ v_x(T/2) \end{pmatrix} = \boldsymbol{0}.$$

If these constraints are satisfied, the trajectory crosses the x–axis orthogonally after half a period, and the symmetry of the PCR3BP can be used to recover the full periodic orbit.

The Jacobian required for the differential correction step in Eq. (4.1) contains the sensitivities of the final values $y(T/2)$ and $v_x(T/2)$ with respect to the initial velocity $v_{y,0}$ and the propagation time $T/2$:

$$D\boldsymbol{F}(\boldsymbol{X}) = \begin{pmatrix} \dfrac{\partial y_f}{\partial v_{y,0}} & \dfrac{\partial y_f}{\partial (T/2)} \\ \dfrac{\partial v_{x,f}}{\partial v_{y,0}} & \dfrac{\partial v_{x,f}}{\partial (T/2)} \end{pmatrix}.$$

The first column is obtained from the entries in rows 2 and 3 of column 4 of the state transition matrix

$$\boldsymbol{\Phi}(T/2, \boldsymbol{x}_0) = \frac{\partial \boldsymbol{\phi}(T/2, \boldsymbol{x}_0)}{\partial \boldsymbol{x}_0},$$

where the state ordering is $\boldsymbol{x} = (x, y, v_x, v_y)$. The second column is given by elements 2 and 3 of the vector field evaluated at the final state:

$$\frac{\partial \boldsymbol{\phi}(T/2, \boldsymbol{x}_0)}{\partial (T/2)} = \boldsymbol{f}(\boldsymbol{x}_f).$$

To ensure convergence of the Newton iteration, a high-quality initial guess $\boldsymbol{X}^{(0)}$ is required. For resonant orbits, this initial guess can be constructed from two-body dynamics. Using the vis-viva equation from two-body orbital mechanics [BMW71], the initial velocity is approximated by

$$v_{y,0}^{(0)} = \pm\sqrt{2(1-\mu)\left(\frac{1}{\rho_1} - \frac{1}{2a}\right)} - \rho_1, \tag{4.5}$$

where μ_1 is the gravitational parameter of the primary, $\rho_1 = x_0 + \mu$ is the initial distance to the primary, and $a = (q/p)^{2/3}$ is the approximate semimajor axis associated with the $p : q$ resonance. An initial guess for the full orbital period is

$$T^{(0)} = 2\pi q,$$

so the corresponding half-period guess used in the differential corrector is $T^{(0)}/2$.

Because the two-body approximation neglects the gravitational perturbation of the secondary, the corrected PCR3BP period is not generally exactly equal to $T^{(0)}$, but it typically remains close. After solving Eq. (4.1) using the Jacobian $D\boldsymbol{F}$, the free variables are updated according to Eq. (4.2). This iteration is repeated until the symmetry constraints are satisfied to the prescribed tolerance. After convergence, the orbit is verified by checking that it has the desired resonant structure, namely the correct number of periapsis or apoapsis passages associated with the chosen resonance ratio $p : q$.

For improved numerical convergence, especially for higher-order resonances, we also employ differential correction with a multiple-shooting rather than a single-shooting transcription. While this is not explained in detail here, the essential idea is to divide the trajectory into several shorter segments, correct the intermediate node states and segment times, and enforce continuity between adjacent segments in addition to the final symmetry conditions.

4.4 Computation of Lyapunov Orbits

We briefly describe how to use the differential correction algorithm, to compute Lyapunov orbits with a given energy $\hat{C}$, since these are used in the following section. This extends naturally from the computation of resonant orbits by relaxing the fixed initial position and constraining the energy value according to

$$\boldsymbol{X} = \begin{pmatrix} x_0 \\ v_{y,0} \\ T/2 \end{pmatrix}, \qquad \boldsymbol{F}(\boldsymbol{X}) = \begin{pmatrix} y(T/2) \\ v_x(T/2) \\ C(x_0, v_{y,0}) - \hat{C} \end{pmatrix} = \mathbf{0}.$$

The Jacobian matrix $D\boldsymbol{F}(\boldsymbol{X})$ is also expanded accordingly. The initial guess is obtained from the linearised dynamics near the collinear libration point by applying a small displacement in x and choosing the corresponding v_y from the centre-mode frequency and eigenvector relation. By the same iterative procedure described above, the corrector then identifies an initial state and period leading to a periodic orbit of the targeted energy level.

5 Invariant Manifolds

The PCR3BP admits the existence of several dynamical structures, such as the Hills regions and zero-velocity curves described in Section 3.4. Among them are *invariant manifolds*, "tubes" that allow low-energy transport between different regions of the system's domain. These manifolds, first traced by Koon et al. in [KLMR00] with the advent of robust numerical methods, can be used to aid in preliminary mission design for cislunar or interplanetary missions. In this section, we will focus on the manifolds associated with the planar Lyapunov $\mathcal{L}_1$ bounded orbits described in Section 4. These are not unique: for every unstable periodic orbit admitted by the PCR3BP over the continuous range of values of the Jacobi constant C, there exists a pair of manifolds associated with the orbit at that energy level [KLMR00]. Thus, an infinite number of such manifolds exists.

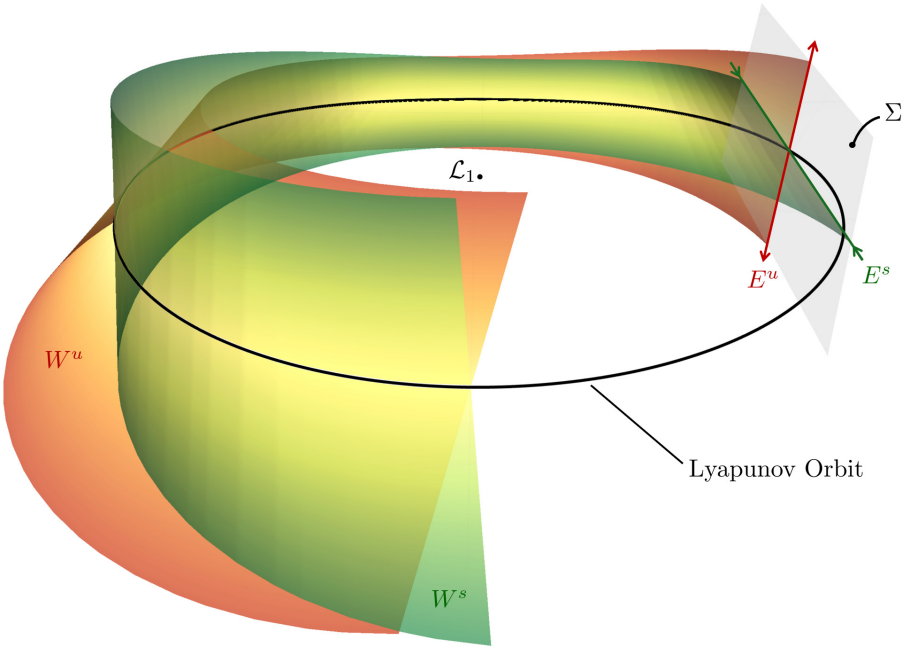


Figure 7: Illustration of stable and unstable subspaces E^s, E^u (depicted on a cross section Σ) and manifolds W^s, W^u for an $\mathcal{L}_1$ planar Lyapunov periodic orbit, plotted in (x, y, v_y) space with the v_y component as the vertical.

The first step in tracing out invariant manifolds is to identify an unstable hyperbolic periodic orbit of a fixed, "target" Jacobi constant energy, $\hat{C}$. We do this in accordance with the process described in Section 4.4. Once the periodic orbit has been computed, provided it is hyperbolic and unstable, its invariant manifolds can be traced. We denote the STM evaluated over one full orbital period, T, (on the PCR3BP domain, where $\boldsymbol{x} = (x, y, v_x, v_y)$) as the *monodromy matrix*, $\boldsymbol{M} = \boldsymbol{M}(\boldsymbol{x})$, written as:

$$\boldsymbol{M}(\boldsymbol{x}) \coloneqq \boldsymbol{\Phi}(T, \boldsymbol{x}_0).$$

The eigenvalues of the monodromy matrix occur in reciprocal pairs (see Section 8 for detail). For a planar periodic orbit, we consider the four eigenvalues and eigenvectors of $\boldsymbol{M}$ in the PCR3BP. One pair of such eigenvalues is unity; the eigenvectors associated with this pair correspond to the orbit's tangent and normal directions. If the orbit is unstable, the remaining two eigenvalues form the reciprocal pair $\{\lambda, 1/\lambda\}$; for practical Earth–Moon orbits, the value of λ is such that $\lambda \gg 1, 1/\lambda \ll 1$ (typically on the order of magnitude $\lambda \sim 10^3$, see Section 8 for detail). The eigenvectors $\boldsymbol{e}^s(\boldsymbol{x}), \boldsymbol{e}^u(\boldsymbol{x})$ of $\boldsymbol{M}$ associated with this pair span the stable and unstable subspaces, which we denote E^s and E^u, respectively.

According to the stable manifold theorem for flows (from our MAE 541 course), the existence of the subspaces implies the existence of stable and unstable manifolds, denoted as W^s and W^u, tangent to these subspaces. The local subspaces and manifolds are visualised for a planar Lyapunov $\mathcal{L}_1$ periodic orbit in Fig. 7. In this figure, the stable and unstable subspaces are plotted on a perpendicular cross-section Σ to the Lyapunov planar orbit, and local portions of the manifolds are computed numerically and plotted. The visualisation plots the orbit on the (x, y) plane, with the v_y component of velocity out of plane.

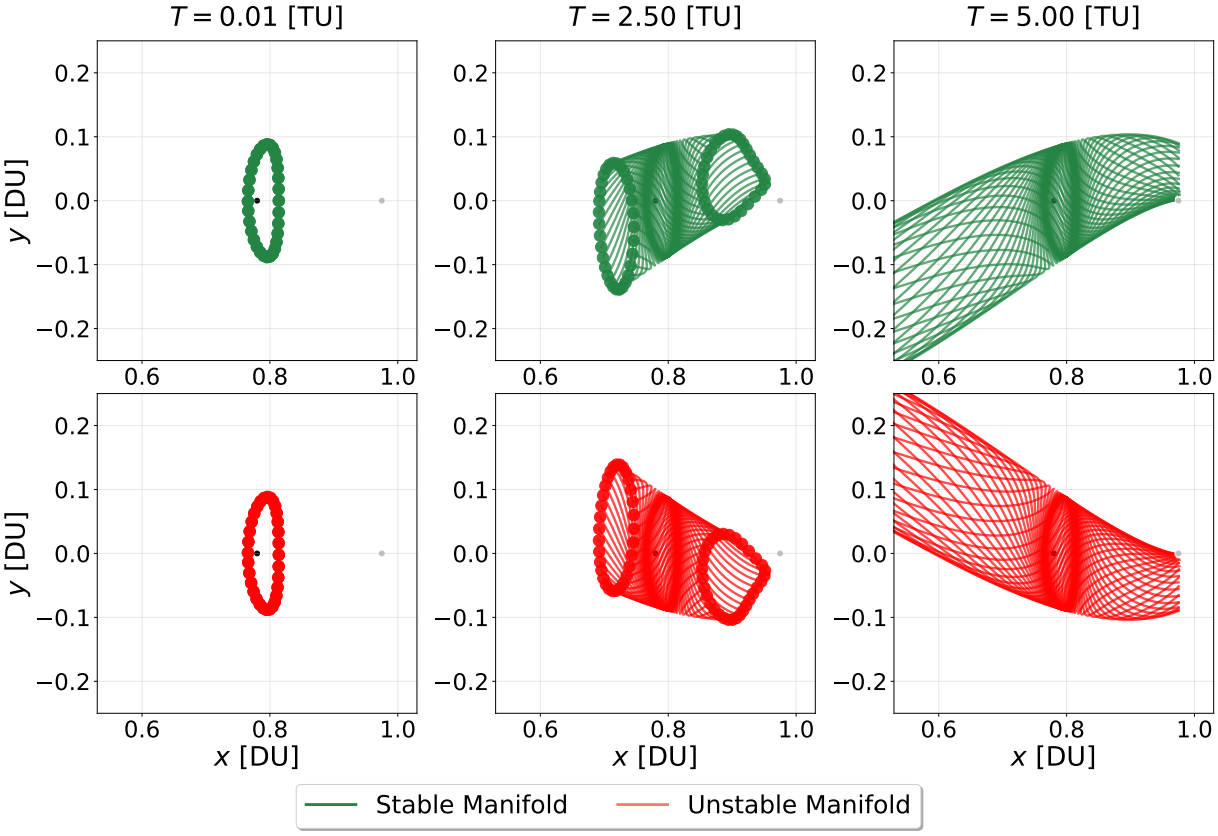
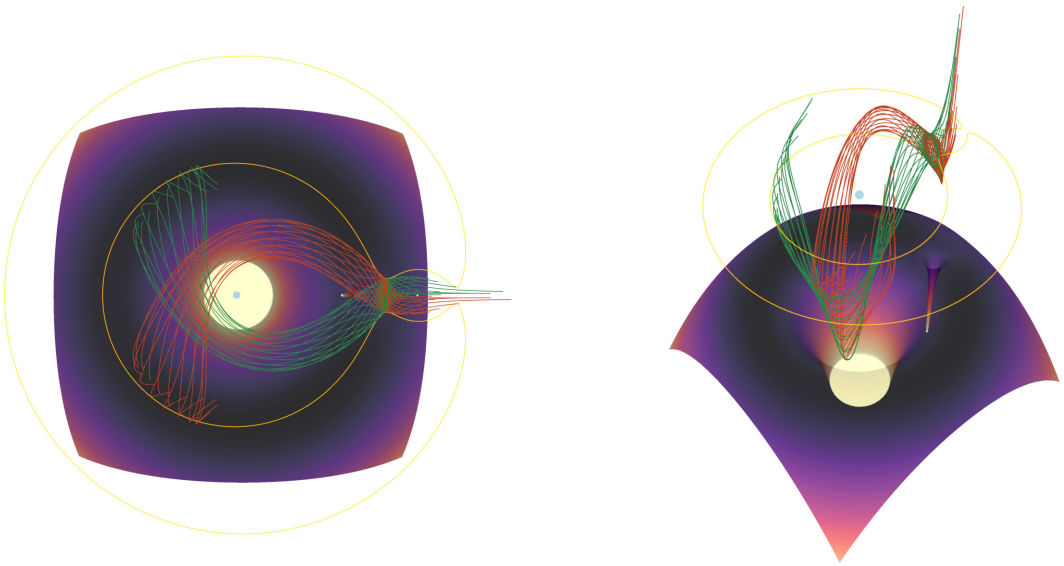


Figure 8: Visualisation of the computation of stable and unstable manifolds across propagation intervals over the (x, y) plane.

To numerically trace the manifolds, we perturb a collection of N initial states chosen at discrete intervals along the periodic orbit, $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^N$, by a small amount ε along the desired subspace. Thus, we generate a series of initial conditions $\boldsymbol{x}_0^{(i),s}$ and $\boldsymbol{x}_0^{(i),u}$, according to

$$\begin{aligned} \boldsymbol{x}_0^{(i),s} &= \boldsymbol{x}_0^{(i)} \pm \varepsilon \boldsymbol{e}^s(\boldsymbol{x}), \quad i \in 1, 2, \ldots, N \\ \boldsymbol{x}_0^{(i),u} &= \boldsymbol{x}_0^{(i)} \pm \varepsilon \boldsymbol{e}^u(\boldsymbol{x}), \quad i \in 1, 2, \ldots, N \end{aligned} \tag{5.1}$$

where ε is chosen on the order of $\varepsilon \sim 10^{-5}$DU. The sign of the perturbation ($\pm$) dictates which branch of the manifold is traced. The perturbed unstable states $\boldsymbol{x}_0^{(i),u}$ are then propagated forwards in time to trace out the unstable manifold W^u, and the stable states $\boldsymbol{x}_0^{(i),s}$ are propagated backwards in time to trace the stable manifold W^s, according to the procedure described in Section 4. The computation of stable and unstable manifolds of the $\mathcal{L}_1$ Lyapunov planar periodic orbit is visualised in Fig. 8. We selected $N = 16$ initial conditions according to Eq. (5.1) for each manifold, and propagated them in both directions for a period of $T = 2.65$ TU using `pydylan`. For ease of visualisation, this and the following figure were numerically computed using a higher mass ratio $\mu = 0.025$ compared to that of the Earth–Moon system.



(a) Two-dimensional view. (b) Three-dimensional view.

Figure 9: Illustration of Lyapunov $\mathcal{L}_1$ invariant manifolds in the PCR3BP.

The finalised manifolds are visualised in both two (left) and three (right) dimensions in Fig. 9, as green and red "tubes", respectively. In this figure, the manifolds are plotted over the (x, y) axis, with the v_y component of velocity out of plane. We also plot the effective potential surface $\tilde{U}$ (purple), the Hill's region boundary (dark yellow), and the primary (light blue dot) and secondary (white dot). The manifolds, Hill's region, and periodic orbit are all associated with the same value of the Jacobi constant $\hat{C}$, where we chose $C_2 > \hat{C} > C_3$ to illustrate a scenario where both the $\mathcal{L}_1$ and $\mathcal{L}_2$ gateways have opened, but the $\mathcal{L}_3$ point is still inaccessible. From Fig. 9b, it is interesting to note that the out-of-plane evolution of the manifolds follows the shape of the effective potential. This makes intuitive sense, as the velocity of smallest body m will increase in proximity to primary and secondary, which corresponds to higher magnitudes of the effective potential. These and other three-dimensional visualisations were made using the `python` library `plotly`; source code for all interactive visualisations and videos can be found at this repository.

6 Poincaré Section

6.1 Initial Reductions for Visualisation

As the points in our state space are six-dimensional vectors of the form of Eq. (2.3), we first apply several considerations to make our investigations more manageable.

The first of these is the least restrictive: As noted in Section 2, the dynamics of the CR3BP are such that the z and v_z components of a point in our state space may be taken to be 0. As such, up to isomorphism/omission of these constantly zero indices, we may think of our space as a subset of $\mathbb{R}^4$, giving the first step in our reduction. This first reduction yields the PCR3BP. Informally, we may think of this as the identification

$$(x, y, 0, v_x, v_y, 0) \cong (x, y, v_x, v_y).$$

Following this, we face the difficulty of simplifying the dynamics of our system with our desire to more easily analyse it. Towards a compromise in this matter, we employ the use of a Poincaré map, giving the cross-section of trajectories in our state space, with the $y = 0$ hyperplane. As is well-established (and known from class), such a use of a Poincaré map still allows us to meaningfully consider the dynamics of a system, while also reducing the dimensionality. In our case, this process results in bringing the above four-dimensional set-up to a more manageable three dimensions. Again, informally this is described by

$$(x, y, v_x, v_y) \to (x, 0, v_x, v_y) \cong (x, v_x, v_y).$$

Finally, when visualisation/conceptual understanding of our system is particularly important, we may further reduce our space to a two-dimensional set-up via the projection down to the x, v_y plane. Similar as above, this is thought of as

$$(x, v_x, v_y) \to (x, v_x).$$

The entirety of this reduction is summed up in the following Fig. 10:

$$\mathbb{R}^6 \xrightarrow[z = v_z = 0]{\text{By PCR3BP}} \mathbb{R}^4 \xrightarrow[y = 0]{\text{By Poincaré map}} \mathbb{R}^3 \xrightarrow[(x, v_y) \text{ plane}]{\text{By projecting to}} \mathbb{R}^2$$

Figure 10: Reduction of the state space dimension.

6.2 Constant Energy Regions

As established in Section 3, the equations of motion of the CR3BP are Hamiltonian and autonomous. The Hamiltonian (or total energy of the smallest body in the CR3BP) is defined in Eq. (3.1) for the full six-dimensional state. In this section, however, we consider a reduced-order coordinate vector. For a general point of the form $(x, y, v_x, v_y) \in \mathbb{R}^4$, the total energy $E = H$ of the system reduces to

$$E = \frac{1}{2}\left(v_x^2 + v_y^2\right) + \tilde{U}(x, y), \tag{6.1}$$

where $\tilde{U}(x, y)$ is the effective potential of the PCR3BP, defined in Section 3, and simplified by taking $z = v_z = 0$ (the first step in our above reduction).

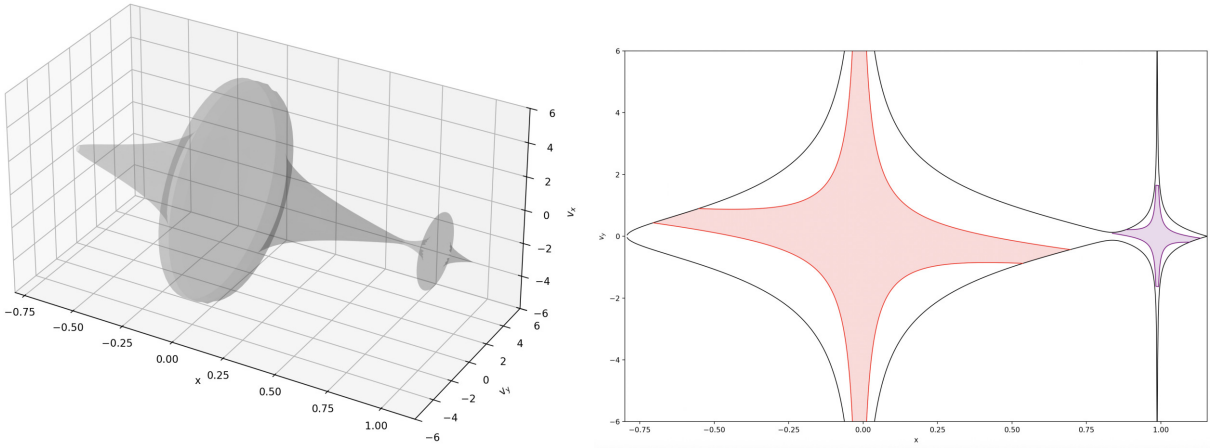


Figure 11: Visualisations of the accessible region in three dimensions and its projection to the (x, v_y) plane, displaying the Earth and Moon crash regions in red and purple, respectively.

With the restriction $y = 0$ in place from our Poincaré section (the second step in our above reduction), and for a given energy value E, it immediately follows from a rearrangement of Eq. (6.1) that all admissible points $(x, 0, v_x, v_y)$ with this energy value are defined by the surface

$$v_x^2 + v_y^2 = 2\big(E + \tilde{U}(x, 0)\big). \tag{6.2}$$

We therefore refer to these surfaces as *surfaces of constant energy*.

One particular such surface is that defined by

$$E_2 \approx -1.5861, \tag{6.3}$$

the energy value (in dimensionless CR3BP units) corresponding to the Lagrange point $\mathcal{L}_2$ (see Table 2 for details). We refer to the region enclosed by this surface as the *accessible region*, as it comprises all states of the form $(x, 0, v_x, v_y)$ that have an energy level at or below E_2. As orbiting bodies with energies above this level may leave the Earth–Moon system, this defines the boundary of the scope of our current work.

As seen in Fig. 11, the plot of the accessible region has two distinct lobes, each roughly shaped like a pseudosphere. The larger of these corresponds to the region around Earth, and the smaller corresponds to that around the Moon. The planar plot also include crash regions around the Earth and the Moon, which are derived in the Appendix in Section C. These regions corresponds to initial conditions that crash into the Earth or the Moon when propagated under the dynamics of the system.

6.3 Orbital Crossings

Periodic orbits in the PCR3BP become fixed points of the Poincaré map at $y = 0$. Following the process laid out in Section 4.3 various families of planar resonant orbits with energy $E < E_2$ were computed in `pydylan`. The equilibrium points corresponding to Poincaré map crossings of these orbits are visualised in Fig. 12, which depicts these crossings in the three-dimensional x, v_x, v_y space, as well as in the projection down to the (x, v_y) plane, for visual clarity. Crossings with both positive and negative v_y are included to provide a complete view, although a single crossing direction would be sufficient under the standard

definition of a Poincaré section. The colours specified in the legend are consistent across both plots. Although it appears as though some points enter the Earth crash region in the 2D image, this is simply a consequence/limitation of the use of projection.

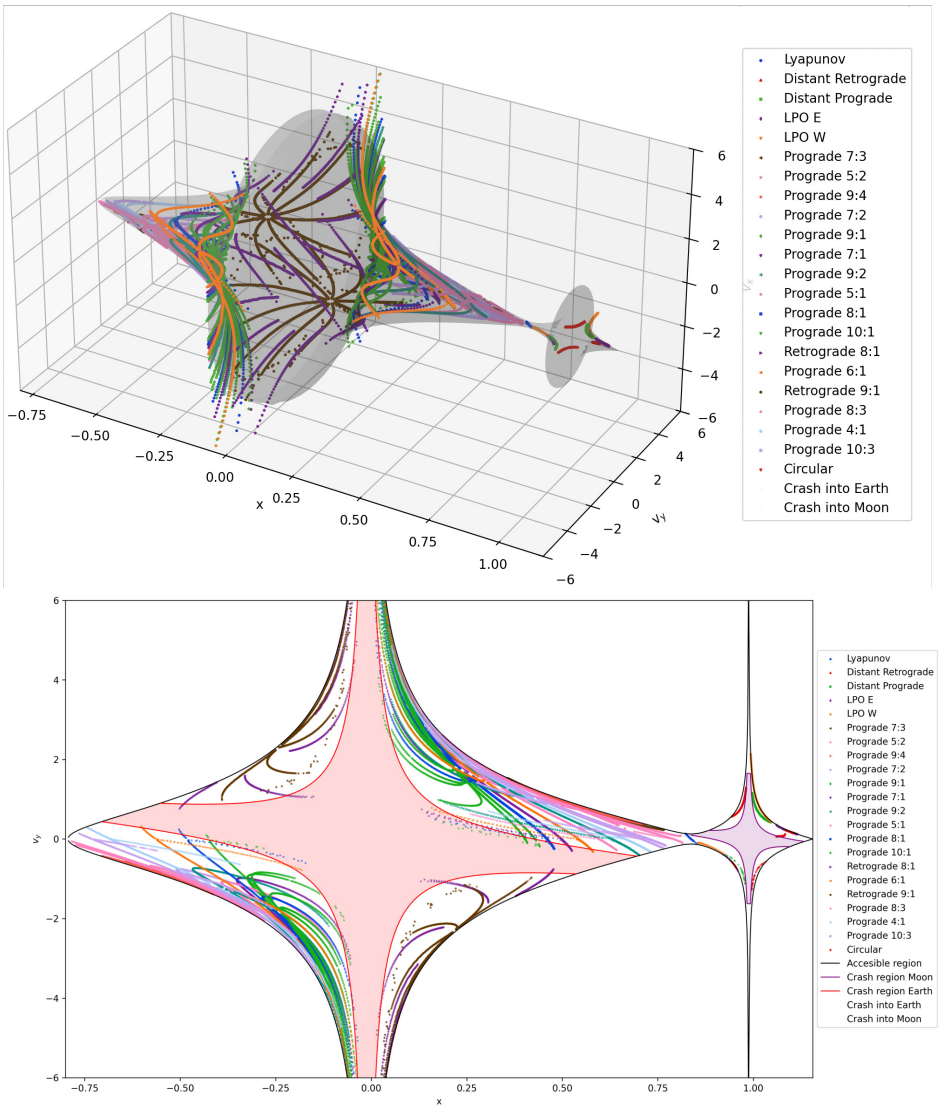


Figure 12: Orbital crossings of several planar resonant orbits, depicted in three-dimensions and the projection to the (x, v_y) plane.

Ultimately, an interesting direction for future work may be investigating the 'coverage' of these crossings within our accessible region, potentially finding a method to cover the space with such points to an arbitrary degree.

Such an effort is not unfounded, results by Poincaré in [Poi93] following a conjecture in [Poi92], demonstrate the density of periodic orbits at least in analogous, low-energy regimes. It has not been ruled out, and is viewed as likely by the authors and by a larger portion of the astrodynamics community, that this density of periodic orbits may extend to our domain as well.

Regardless, as can be seen in Fig. 12, at least with the families of orbits tested, we have struggled to find families with many crossings close to the inner portion of the space/near the crash region. (Qualitatively, one can see that the points plotted are much more 'dense'

near the boundary of the accessible region. This is at least true when viewing the upper right and lower left quadrants, as these house the crossings of prograde orbits, which are much more numerous in our tested families due to data availability/the data we computed. Not much should be read into the sparsity of points in the upper left and lower right quadrants (and lunar region), for this reason.)

7 Fixed-Energy Poincaré Maps

Another way to visualise the dynamics of the system and the role of periodic orbits in a two-dimensional representation is to again consider a Poincaré section at $y = 0$, but now with the energy fixed to a prescribed value. This additional restriction reduces the section to a two-dimensional surface embedded in the three variables (x, v_x, v_y), such that any two variables can be used for visualisation while the third is determined by the specified energy level, using Eq. (6.1). The (x, v_x) projection is shown in Fig. 13 for several values of the

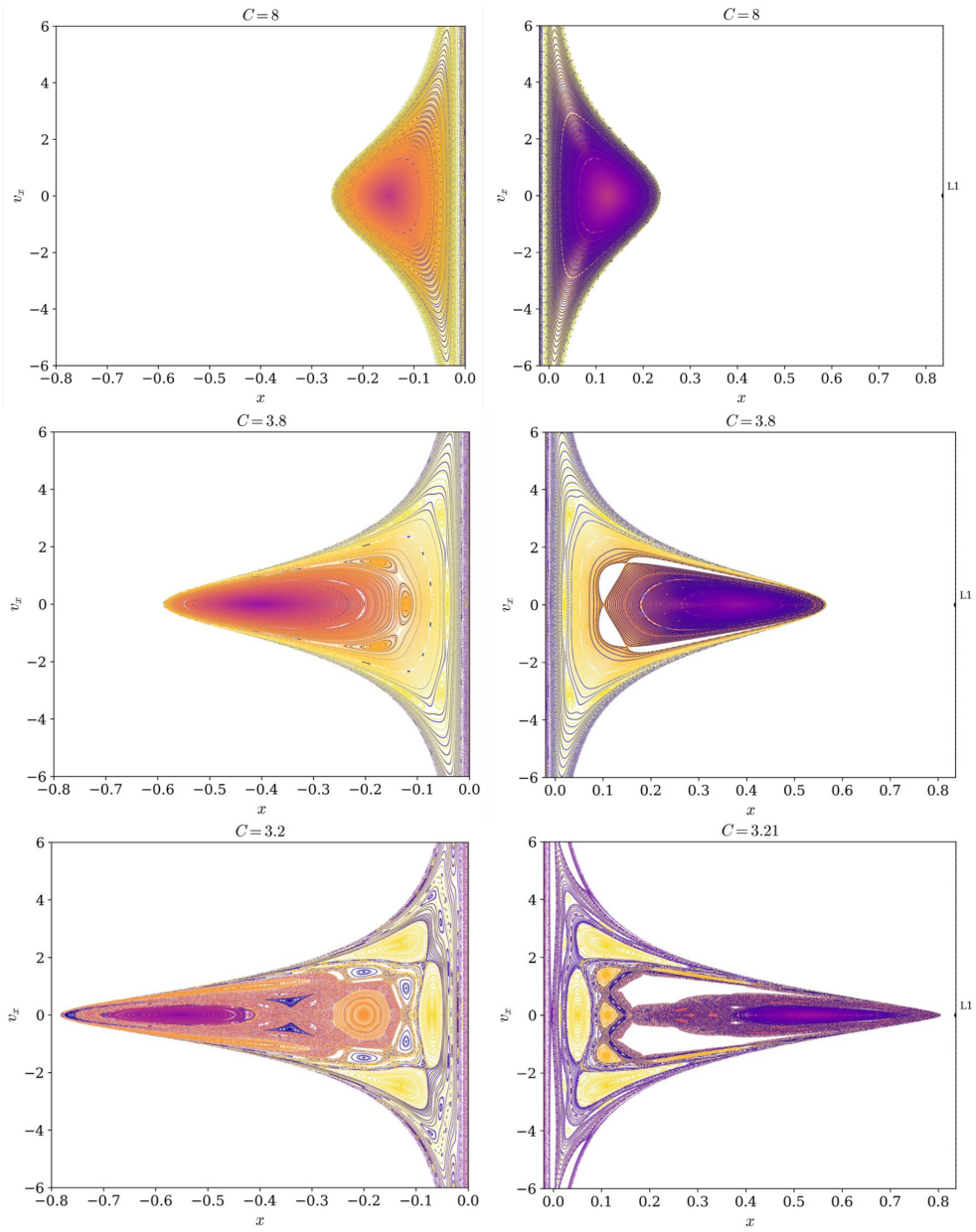


Figure 13: Poincaré sections for different Jacobi constants.

Jacobi constant C, which is equivalent to fixing the energy, as shown in Eq. (3.7).

Unlike in the previous section, we do not only overlay periodic orbits on the section. Instead, we propagate a large number of initial conditions to illustrate the chaotic structure of the phase space. To generate these plots, 800 evenly spaced initial conditions were placed along the x–axis, with $y = 0$ and $v_x = 0$, within the energetically accessible range for each prescribed value of C. The corresponding value of v_y was then computed so that each initial condition lies on the selected energy level. Each trajectory was propagated under the CR3BP dynamics until 3000 crossings of the Poincaré section at $y = 0$ were recorded. In exact arithmetic, all crossing points would remain on the same energy surface as the initial condition. Using the adaptive step-size RK54 integrator employed here, the maximum numerical energy drift over the considered propagation time is 10^{-6}. In Fig. 13, each crossing point is colour-coded according to the x-coordinate of its initial condition, providing a visual indication of how trajectories move through phase space.

For low energy levels, such as $C = 8$, the accessible region is relatively small and the motion remains largely ordered, evolving around an island associated with a stable resonant orbit. As the energy increases, as illustrated for $C = 3.8$, additional islands appear, each corresponding to a stable resonant orbit. For $C = 3.2$, which lies just below E_2 (recall Eq. (3.7) and Eq. (6.3)), the chaotic nature of the system becomes increasingly apparent. The resonant orbits associated with many of the stability islands at this energy level are shown in Fig. 14. As the figure illustrates, these islands are characterised by concentric rings centred on stable resonant orbits. Between the islands, several regions exhibit clearly chaotic behaviour, including strong mixing between trajectories originating from different initial conditions. These chaotic regions likely contain transversal intersections of stable and unstable invariant manifolds associated with unstable resonant orbits.

In fact, the realisation communicated in Fig. 14 that certain resonant orbits lie in relatively more stable regions, is one that plays a large role in spacecraft trajectory design. The observed phenomenon relates to a more general study of *lobe sequences/dynamics*, and can be studied by analysing heteroclinic tangles of the relevant Poincaré map [HBSH26].

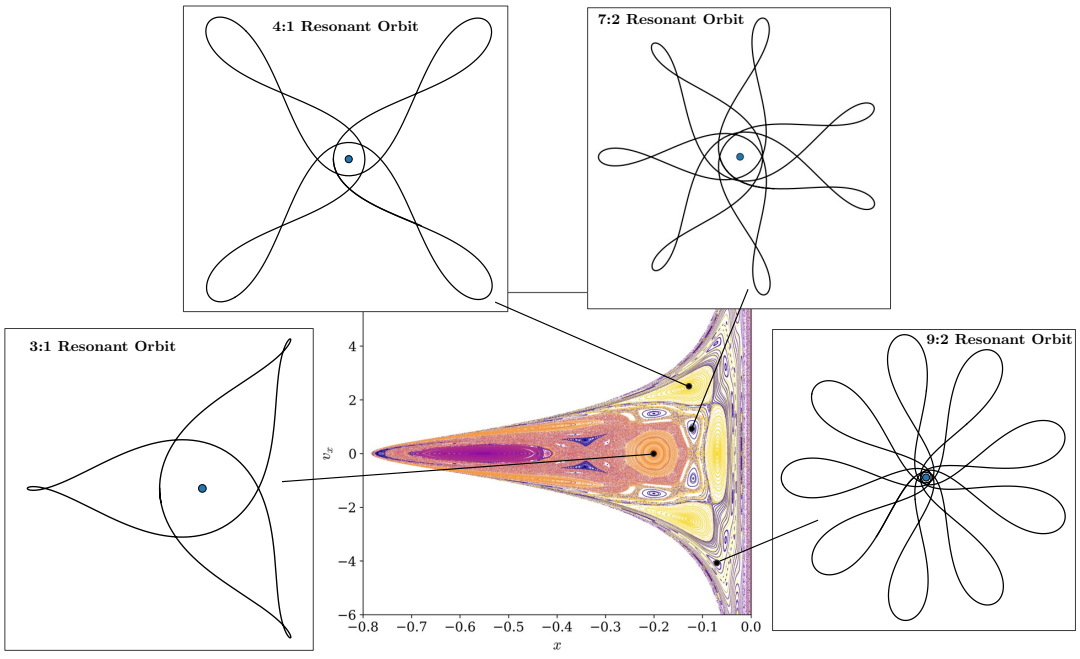


Figure 14: Depiction of several stable points in the $C = 3.2$ diagram corresponding to low ratio $p : q$ resonant orbits.

8 Normal Forms Around Periodic Orbits

Normal form theory provides a systematic way to simplify a non-linear dynamical system near an invariant object through a suitable change of coordinates. The central idea is to apply a sequence of near-identity transformations that remove non-essential non-linear terms while retaining the dominant structure of the dynamics. For Hamiltonian systems, such as the CR3BP, these transformations must be symplectic so that the Hamiltonian structure of the original system is preserved. The resulting simplified system is advantageous because its dynamics can be described analytically, enabling closed-form propagation in the transformed coordinates. As a result, the long-term evolution of objects near a periodic orbit can be approximated in a single evaluation rather than through computationally expensive numerical integration over many years or decades. This capability is particularly useful for applications such as predicting the evolution of space debris in the cislunar domain over long time-horizons. In the following, we provide an overview of normal form theory in the context of periodic orbits in the CR3BP and present numerical results demonstrating its application.

8.1 Normal Form Theory

This section provides only a concise overview of the normal form transformations used in this work; a more complete theoretical description can be found in Sinha's dissertation [Sin26]. The methodology follows the numerical normal form framework developed by Jorba and Villanueva [JV98, Jor99]. The implementation used in the present analysis was developed with contributions from Sinha [Sin26] and is applied here to the examples considered below.

We describe the transformations at a high level and emphasise how the Hamiltonian changes at each step. As detailed in Section 3, the CR3BP satisfies the Hamilton canonical equation Eq. (3.3) with Hamiltonian H defined in Eq. (3.1). In this section, we adopt the canonical coordinates $\boldsymbol{z} = (\boldsymbol{r}, \boldsymbol{p})$, where $\boldsymbol{p}$ is the canonical momentum (see Section 3 for definitions). The coordinate transformations used below are *symplectic*, meaning that they preserve this canonical Hamiltonian structure.

Let the periodic orbit be denoted by $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^6$, with $\boldsymbol{\gamma}(t+T) = \boldsymbol{\gamma}(t)$. The first step in constructing the normal form is to translate the periodic orbit to the origin using

$$\boldsymbol{\xi} = \boldsymbol{z} - \boldsymbol{\gamma}(t). \tag{8.1}$$

The Hamiltonian in the translated coordinates, denoted by $K_1(t, \boldsymbol{\xi})$, is time dependent. Expanding about $\boldsymbol{\xi} = 0$ gives

$$K_1(t, \boldsymbol{\xi}) = \frac{1}{2}\boldsymbol{\xi}^T \nabla^2 H(\boldsymbol{\gamma}(t))\boldsymbol{\xi} + \sum_{d \geq 3} H_d^{(\boldsymbol{\xi})}(t, \boldsymbol{\xi}), \tag{8.2}$$

where $\nabla^2 H(\boldsymbol{\gamma}(t))$ is the Hessian of the original Hamiltonian evaluated along the periodic orbit, and $H_d^{(\boldsymbol{\xi})}$ denotes the homogeneous polynomial terms of degree d in $\boldsymbol{\xi}$. The constant and linear terms vanish because the expansion is taken about the translated periodic solution.

The next transformation removes the explicit time dependence from the quadratic part of the Hamiltonian. This is achieved through a Floquet transformation to new coordinates $\boldsymbol{\eta}$,

$$\boldsymbol{\eta} = \boldsymbol{P}^{-1}(t)\boldsymbol{\xi}. \tag{8.3}$$

Here, $\boldsymbol{P}(t)$ is the periodic part of the STM decomposition

$$\boldsymbol{\Phi}(t) = \boldsymbol{P}(t)e^{t\boldsymbol{B}}, \tag{8.4}$$

where $\boldsymbol{P}(t+T) = \boldsymbol{P}(t)$ and $\boldsymbol{P}(0) = \boldsymbol{I}_6$. The matrix $\boldsymbol{\Phi}(t)$ is the STM of the variational dynamics in the translated $\boldsymbol{\xi}$ coordinates, defined in the same sense as Eq. (4.3) and $\boldsymbol{B}$ is a Floquet exponent matrix. Once $\boldsymbol{B}$ is determined from

$$\boldsymbol{B} = \frac{1}{T}\log(\boldsymbol{\Phi}(T))\,, \tag{8.5}$$

the periodic matrix $\boldsymbol{P}(t)$ follows from Eq. (8.4). In Floquet coordinates, the Hamiltonian takes the form

$$K_2(t,\boldsymbol{\eta}) = \frac{1}{2}\boldsymbol{\eta}^\top \boldsymbol{S}\boldsymbol{\eta} + \sum_{d\geq 3} H_d^{(\boldsymbol{\eta})}(t,\boldsymbol{\eta}), \tag{8.6}$$

where the quadratic part is now autonomous and $\boldsymbol{S} = -\boldsymbol{J}\boldsymbol{B}$, and $\boldsymbol{J}$ is the symplectic matrix introduced in Section 3.

The remaining transformations remove higher-order nonresonant terms from the Hamiltonian. At each degree d, this is done by solving the homological equation

$$\mathcal{L}_d\tilde{W}_d + Z_d = H_d^{(\boldsymbol{\eta})}. \tag{8.7}$$

Here, $\tilde{W}_d(t,\boldsymbol{\eta})$ is the generating function of the near-identity transformation, and Z_d contains the terms that remain in the normal form. Ideally, $\tilde{W}_d$ is chosen so that $Z_d = 0$; however, terms that lie in the kernel of the homological operator cannot be removed and are retained as resonant terms. The homological operator is defined as

$$\mathcal{L}_d\tilde{W} \equiv \partial_t\tilde{W} + \{\tilde{W}, H_2\}, \qquad \{\tilde{W}, H_2\} \equiv \nabla\tilde{W}^\top J\nabla H_2, \tag{8.8}$$

where $H_2 = \frac{1}{2}\boldsymbol{\eta}^\top\boldsymbol{S}\boldsymbol{\eta}$ is the quadratic Hamiltonian.

Since the higher-order terms remain time periodic, they can be expanded in a Fourier series,

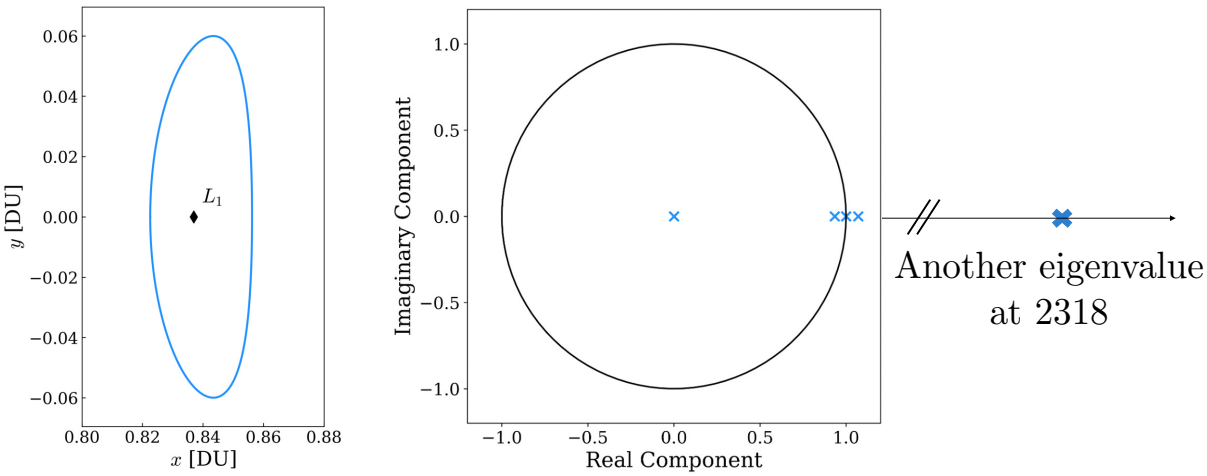
$$H_d^{(\boldsymbol{\eta})}(t,\boldsymbol{\eta}) = \sum_{|k|\leq K} H_{d,k}(\boldsymbol{\eta})e^{ik\omega t}, \qquad \omega = \frac{2\pi}{T}, \tag{8.9}$$

where the series is truncated at Fourier order K. Expanding $\tilde{W}_d$ and Z_d in the same form decouples the homological equation by polynomial degree and Fourier mode, allowing the normal form transformation to be computed term by term.

The final Hamiltonian consists of the autonomous quadratic part together with the resonant higher-order terms that cannot be removed. It defines the approximate dynamics in the normal form coordinates $\boldsymbol{u}$, in which objects can be propagated analytically.

8.2 Numerical Results

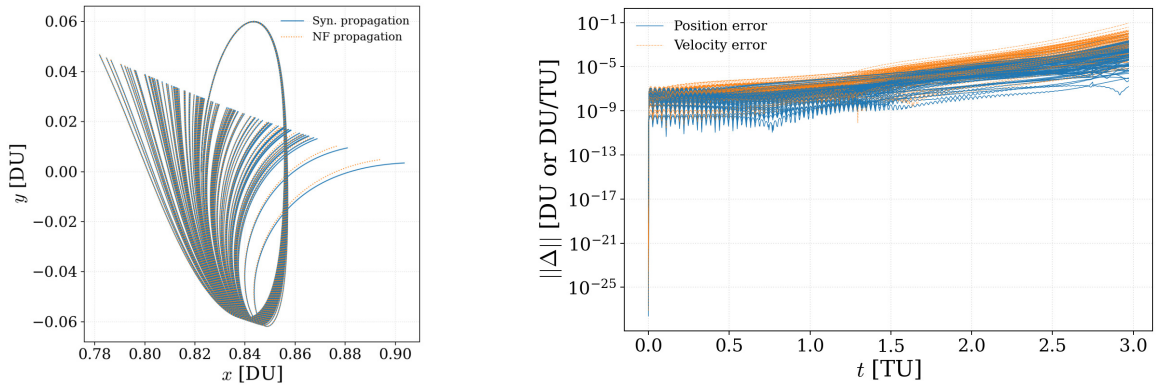
We test the normal form construction using a planar Lyapunov orbit around the libration point $\mathcal{L}_1$. The orbit is shown in Fig. 15 together with its Floquet multipliers, defined as the eigenvalues of the monodromy matrix. Because the orbit lies in a region where the gravitational influence of both the Earth and the Moon is significant, its dynamics are highly unstable, as indicated by the multiplier far outside the unit circle. This instability makes the orbit a challenging test case for the normal form approximation.



(a) Trajectory in (x, y). (b) Floquet multipliers in the complex plane.

Figure 15: Planar Lyapunov orbit used to test the normal form construction.

To demonstrate the application of the normal form, we initialise a set of nearby trajectories from a Gaussian distribution centred on the Lyapunov orbit and propagate them using both the full synodic CR3BP dynamics and the simplified normal form system. The initial Gaussian is sampled on the Poincaré section $y = 0$ around the initial crossing point of the Lyapunov orbit. The standard deviations are chosen as $\sigma_x = 10^{-6}\,\mathrm{DU}\,(384\,\mathrm{m})$, $\sigma_{v_x} = 10^{-4}\,\mathrm{VU}\,(0.1\,\mathrm{m/s})$, and $\sigma_{v_y} = 10^{-4}\,\mathrm{VU}\,(0.1\,\mathrm{m/s})$. We draw 100 samples from this distribution and map each sample to normal form coordinates using the transformations $\boldsymbol{z} \to \boldsymbol{u}$ described above. To limit the computational cost, the normal form is truncated at polynomial degree $D = 4$, and the time-periodic coefficients are represented using a Fourier expansion of order $K = 16$. The samples are then propagated analytically in normal form coordinates for one orbital period. For comparison, the same initial conditions are propagated in the synodic CR3BP coordinates using an adaptive step-size RK54 integrator. To compare both propagation methods, the analytically propagated normal form trajectories are mapped back to the synodic frame through the inverse transformation $\boldsymbol{u} \to \boldsymbol{z}$ at the same time instances used by the numerical integrator. The results are shown in Fig. 16.



(a) Trajectories in (x, y) space. (b) Position and velocity errors.

Figure 16: Comparison of propagation in synodic and normal form coordinates.

The left panel of Fig. 16 compares the trajectories obtained with both propagation methods in the (x, y) plane. Despite the small initial perturbations, the trajectories depart rapidly from the reference orbit over one period due to the strong instability of the Lyapunov orbit.

Nevertheless, the normal form propagation closely follows the full synodic propagation while the trajectories remain near the reference orbit. As the trajectories move farther away, the accuracy decreases because the normal form is a local approximation; however, it still captures the qualitative behaviour of the perturbed trajectories. This behaviour is also reflected in the right panel, which shows the $\mathcal{L}_2$ norms of the position and velocity errors over time for all propagated trajectories.

9 Conclusion

As evidenced by the content throughout this paper, the CR3BP defines a dynamical system rich in structure and is one that has captivated mathematical minds over the span of centuries. Even today (*especially* today), the Earth-Moon CR3BP remains an important model for the modern study of astrodynamics.

In spite of its simplified assumptions, the model provides a prime framework for us to study. It continues to be a key tool for comprehending the dynamics of orbital bodies in our neighbourhood of the Solar System, be they spacecraft, orbital debris, or otherwise.

As interest in the cislunar region continues to grow, the population of satellites (and eventually, of space junk) therein will only grow correspondingly. It is for this reason that the need to deepen our understanding of this system, even with its centuries-old history, is still so timely. To do so necessarily means deepening our understanding of the complex mathematical structures underlying it as well. Although far more work remains before this can truly be said to be accomplished, we believe that our above efforts place us on a firm footing.

A Analytical Derivation of System Dynamics

Inertial Frame

We begin by considering the general 3BP, which studies the dynamics of an isolated system containing three point-mass bodies with position vectors $\bar{\boldsymbol{r}}_{(i)} \coloneqq (\bar{x}_{(i)}, \bar{y}_{(i)}, \bar{z}_{(i)})$ and masses $\bar{m}_i$, where $i \in \{1, 2, 3\}$. Combining Newton's Law of Universal Gravitation with his Third Law, the three masses mutually attract each other according to the equation of motion

$$\bar{m}_i \frac{d^2 \bar{\boldsymbol{r}}_{(i)}}{d\bar{t}^2} = G \sum_{j \neq i} \frac{\bar{m}_i \bar{m}_j}{\bar{r}_{(ij)}^3} \bar{\boldsymbol{r}}_{(ij)}.$$

where G is the Gravitational Constant, $\bar{\boldsymbol{r}}_{(ij)} \coloneqq \bar{\boldsymbol{r}}_{(j)} - \bar{\boldsymbol{r}}_{(i)}$ is the relative position of each point-mass $\bar{m}_j$ with respect to $\bar{m}_i$, and $\bar{r}_{(ij)} \coloneqq \|\bar{\boldsymbol{r}}_{(ij)}\|_2^2$ is the magnitude of the relative position vector, with $\|\cdot\|_2^2$ being the Euclidean norm in $\mathbb{R}^3$. To simplify the dynamics, in the CR3BP we impose the following assumptions and notation.

A system of natural units is used to normalise position, time, and velocity variables: the distance unit (DU) is the distance between the primary and secondary, the time unit (TU) is the orbital period of the primary and secondary divided by 2π, and the mass unit (MU) is $\bar{m}_1 + \bar{m}_2$. Substituting the normalised variables $m_i \coloneqq \bar{m}_i/\text{MU}$, $t \coloneqq \bar{t}/\text{TU}$, and $\boldsymbol{r}_{(i)} \coloneqq \bar{\boldsymbol{r}}_{(i)}/\text{DU}$ and rearranging, we obtain

$$m_i \frac{d^2 \boldsymbol{r}_{(i)}}{dt^2} = \left(\frac{G \cdot \text{MU} \cdot \text{TU}^2}{\text{DU}^3} \right) \sum_{j \neq i} \frac{m_i m_j}{r_{(ij)}^3} \boldsymbol{r}_{(ij)}.$$

The constant term is the product of terms $G \cdot \text{MU}/\text{DU}^3$ and TU^2. By Kepler's Third Law and definition of the time unit, we have

$$\text{TU} = \sqrt{\frac{\text{DU}^3}{G \cdot \text{MU}}}.$$

The constant term thus evaluates to unity, yielding the normalised equations of motion.

To simplify the system even further, we assume that the third body has negligible mass, $m_3 \ll m_1, m_2$. Consequently, this body does not exert any force on the other two and does not affect their motion. We refer to the larger bodies as the *primary*, m_1, and the *secondary*, m_2, where $m_1 > m_2$. For brevity, we rename the smallest body's normalised mass and normalised position vector $m = m_3$ and $\boldsymbol{r} = \boldsymbol{r}_{(3)}$. In the inertial frame, the simplified, normalised equation of motion for the smallest body is thus given by

$$\frac{d^2 \boldsymbol{r}}{dt^2} = \frac{d^2 \boldsymbol{r}_{(3)}}{dt^2} = \frac{m_1}{r_{(13)}^3} \boldsymbol{r}_{(13)} + \frac{m_2}{r_{(12)}^3} \boldsymbol{r}_{(12)}. \tag{A.1}$$

We can reduce the number of the system's parameters to just one by defining the *mass ratio* according to Eq. (2.1), or

$$\mu \coloneqq \frac{m_2}{m_1 + m_2}.$$

In the normalised system, the mass of the primary m_1 is thus $1 - \mu$, and that of the secondary m_2 is μ.

Synodic Frame

To complete the set of assumptions behind the CR3BP, we shift the reference frame from the inertial (fixed) frame, described so far, to a *synodic* (rotating) reference frame. This frame considers the position P of the smallest body m.

We denote as (x, y, z) the coordinates of P expressed in the synodic reference frame. We place the origin of our frame at the barycentre of primary and secondary, with the x–axis aligned with the line joining these two bodies, oriented in the direction of the secondary m_2. The frame is then made to rotate, along with the primary and secondary, around the z–axis with constant angular velocity ω equal to the mean motion of the primaries. Given our normalisation, the angular velocity is unitary, and the angular velocity vector is thus given by $\boldsymbol{\omega} = (0, 0, 1)$. The y–axis is set so to complete a right-handed coordinate system.

In this configuration, the primary m_1 and secondary m_2 are fixed on the x–axis at positions $(-\mu, 0, 0)$ and $(1-\mu, 0, 0)$, respectively. The synodic reference frame is illustrated in Fig. 17.

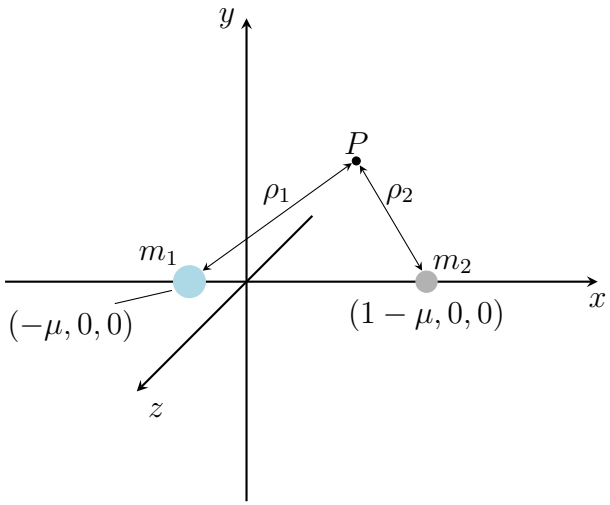


Figure 17: Illustration of the synodic reference frame.

In the synodic frame, the equation of motion of P is augmented following the kinematic transport theorem[2]

$$\left(\frac{d^2\boldsymbol{r}}{dt^2}\right)_I = \left(\frac{d^2\boldsymbol{r}}{dt^2}\right)_R + \underbrace{2\left(\boldsymbol{\omega} \times \left(\frac{d\boldsymbol{r}}{dt}\right)_R\right)}_{\text{Coriolis acceleration}} + \underbrace{\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r})}_{\text{centrifugal acceleration}} + (\dot{\boldsymbol{\omega}} \times \boldsymbol{r})\,. \tag{A.2}$$

[2]Recall that for a rotating reference system, where I denotes the inertial frame and R denotes the synodic frame, the following expressions hold:

$$\begin{aligned}
\left(\frac{d}{dt}\right)_I &= \left(\frac{d}{dt}\right)_R + \boldsymbol{\omega}\times, \\
\left(\frac{d^2\boldsymbol{r}}{dt^2}\right)_I &= \left[\left(\frac{d}{dt}\right)_R + \boldsymbol{\omega}\times\right]\left[\left(\frac{d\boldsymbol{r}}{dt}\right)_R + \boldsymbol{\omega}\times\boldsymbol{r}\right] \\
&= \left(\frac{d}{dt}\right)_R\left[\left(\frac{d\boldsymbol{r}}{dt}\right)_R + \boldsymbol{\omega}\times\boldsymbol{r}\right] + \boldsymbol{\omega}\times\left[\left(\frac{d\boldsymbol{r}}{dt}\right)_R + \boldsymbol{\omega}\times\boldsymbol{r}\right] \\
&= \left(\frac{d^2\boldsymbol{r}}{dt^2}\right)_R + \left(\frac{d\boldsymbol{\omega}}{dt}\right)_R \times \boldsymbol{r} + \boldsymbol{\omega}\times\left(\frac{d\boldsymbol{r}}{dt}\right)_R + \boldsymbol{\omega}\times\left(\frac{d\boldsymbol{r}}{dt}\right)_R + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}) \\
&= \left(\frac{d^2\boldsymbol{r}}{dt^2}\right)_R + \dot{\boldsymbol{\omega}}\times\boldsymbol{r} + 2\boldsymbol{\omega}\times\left(\frac{d\boldsymbol{r}}{dt}\right)_R + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}).
\end{aligned}$$

We remark that the second and third terms on the right-hand side of Eq. (A.2) can be identified with the Coriolis and centrifugal accelerations, respectively. Since angular acceleration ω is constant, the fourth term on the right-hand side of Eq. (A.2) evaluates to zero. We proceed by resolving each of the surviving terms in turn.

Denoting the velocity vector of the smallest body m expressed in the synodic reference frame as $\boldsymbol{v} \coloneqq (d\boldsymbol{r}/dt)_R = (v_x, v_y, v_z)$, the Coriolis acceleration term evaluates to

$$2\begin{pmatrix}0\\0\\1\end{pmatrix} \times \begin{pmatrix}v_x\\v_y\\v_z\end{pmatrix} = 2\begin{pmatrix}-v_y\\v_x\\0\end{pmatrix}.$$

Similarly, the centrifugal acceleration term evaluates to

$$\begin{pmatrix}0\\0\\1\end{pmatrix} \times \left(\begin{pmatrix}0\\0\\1\end{pmatrix} \times \begin{pmatrix}x\\y\\z\end{pmatrix}\right) = \begin{pmatrix}0\\0\\1\end{pmatrix} \times \begin{pmatrix}-y\\x\\0\end{pmatrix} = \begin{pmatrix}-x\\-y\\0\end{pmatrix}.$$

By Newton's Second Law, the left-hand side term in Eq. (A.2) is equal to the sum of external forces per unit mass, and thus equates to the acceleration term given by Eq. (A.1). We simplify notation by expressing the scalar distances from P to each of the other two bodies as $\rho_1 \coloneqq r_{(13)}$ and $\rho_2 \coloneqq r_{(23)}$. Expanding the Euclidean norm and simplifying, we obtain

$$\rho_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \qquad \rho_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}.$$

Substituting these into Eq. (A.1) yields

$$\left(\frac{d^2\boldsymbol{r}}{dt^2}\right)_I = -\frac{1-\mu}{\rho_1^3}\begin{pmatrix}x+\mu\\y\\z\end{pmatrix} - \frac{\mu}{\rho_2^3}\begin{pmatrix}x-1+\mu\\y\\z\end{pmatrix}.$$

Further denoting the acceleration of m expressed in the synodic frame as $(d^2\boldsymbol{r}/dt^2)_R = \ddot{\boldsymbol{r}} = (\ddot{x}, \ddot{y}, \ddot{z})$, and substituting all terms into Eq. (A.2), we arrive at the CR3BP dynamics:

$$\boxed{\ddot{\boldsymbol{r}} = \boldsymbol{g}(\boldsymbol{r}, \boldsymbol{v}) = \begin{pmatrix} 2v_y + x - (1-\mu)\dfrac{x+\mu}{\rho_1^3} - \mu\dfrac{x-1+\mu}{\rho_2^3} \\ -2v_x + y - (1-\mu)\dfrac{y}{\rho_1^3} - \mu\dfrac{y}{\rho_2^3} \\ -(1-\mu)\dfrac{z}{\rho_1^3} - \mu\dfrac{z}{\rho_2^3} \end{pmatrix}}. \tag{A.3}$$

This is a second-order, continuous dynamical system $\boldsymbol{g} : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$. It is worth noting that the system admits the following symmetries:

1. with respect to the (x, y) plane, if $(x(t), y(t), z(t))$ is a solution of Eq. (A.3), then $(x(t), y(t), -z(t))$ is also a solution;
2. with respect to time reversal, if $(x(t), y(t), z(t))$ is a solution of Eq. (A.3), then $(x(-t), -y(-t), z(-t))$ is also a solution;
3. with respect to the mass ratio, if $(x(t), y(t), z(t))$ is a solution of Eq. (A.3) for μ, then $(-x(t), -y(t), z(t))$ is a solution for $1 - \mu$.

B Stability Analysis

To analyse the stability of the Lagrange points, we linearise the second-order equations of motion Eq. (A.3) about each equilibrium point $\mathcal{L}_i$, and study the dynamics on a state $\boldsymbol{x} = \boldsymbol{x}_{\mathcal{L}_i} + \delta\boldsymbol{x}$. Using the second-order system rather than the first-order, six-dimensional system is better suited to the CR3BP for two main reasons. The first is that the three velocity states (v_x, v_y, v_z) only contribute linearly everywhere in the first-order dynamics, and thus the dimensional increase of the Jacobian matrix from $\mathbb{R}^{3\times3}$ to $\mathbb{R}^{6\times6}$ does not reveal any insight in the linearisation. The second reason is that, since the second-order formulation directly follows from the application of Newton's Laws (as detailed in Section A), this approach better highlights the role of the effective potential $\tilde{U}$ and separates the Coriolis term's contribution, as will be made clear in the following analysis. Consider the perturbed position vector

$$\boldsymbol{x} = \boldsymbol{x}_{\mathcal{L}_i} + \delta\boldsymbol{x} = \big(x_{\mathcal{L}_i} + \delta x,\, y_{\mathcal{L}_i} + \delta y,\, z_{\mathcal{L}_i} + \delta z\big).$$

Applying the equations of motion Eq. (A.3) to $\boldsymbol{x}$ and expanding $\tilde{U}$ about $\mathcal{L}_i$ gives

$$\begin{aligned}
\ddot{x}_{\mathcal{L}_i} - 2\dot{y}_{\mathcal{L}_i} + \delta\ddot{x} - 2\delta\dot{y} &= \frac{\partial\tilde{U}}{\partial x}(\boldsymbol{x}) = \frac{\partial\tilde{U}}{\partial x}(\mathcal{L}_i) + \delta x\frac{\partial^2\tilde{U}}{\partial x\partial x}(\mathcal{L}_i) + \delta y\frac{\partial^2\tilde{U}}{\partial x\partial y}(\mathcal{L}_i) + \delta z\frac{\partial^2\tilde{U}}{\partial x\partial z}(\mathcal{L}_i),\\
\ddot{y}_{\mathcal{L}_i} + 2\dot{x}_{\mathcal{L}_i} + \delta\ddot{y} + 2\delta\dot{x} &= \frac{\partial\tilde{U}}{\partial y}(\boldsymbol{x}) = \frac{\partial\tilde{U}}{\partial y}(\mathcal{L}_i) + \delta x\frac{\partial^2\tilde{U}}{\partial y\partial x}(\mathcal{L}_i) + \delta y\frac{\partial^2\tilde{U}}{\partial y\partial y}(\mathcal{L}_i) + \delta z\frac{\partial^2\tilde{U}}{\partial y\partial z}(\mathcal{L}_i),\\
\ddot{z}_{\mathcal{L}_i} + \delta\ddot{z} &= \frac{\partial\tilde{U}}{\partial z}(\boldsymbol{x}) = \frac{\partial\tilde{U}}{\partial z}(\mathcal{L}_i) + \delta x\frac{\partial^2\tilde{U}}{\partial z\partial x}(\mathcal{L}_i) + \delta y\frac{\partial^2\tilde{U}}{\partial z\partial y}(\mathcal{L}_i) + \delta z\frac{\partial^2\tilde{U}}{\partial z\partial z}(\mathcal{L}_i),
\end{aligned}$$

where the first-order terms on the left-hand side represent the Coriolis acceleration. The second derivatives of $\tilde{U}$ are

$$\begin{aligned}
\tilde{U}_{xx} &\equiv \frac{\partial^2\tilde{U}}{\partial x^2} = 1 - \frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3} + 3(1-\mu)\frac{(x+\mu)^2}{\rho_1^5} + 3\mu\frac{(x-1+\mu)^2}{\rho_2^5},\\
\tilde{U}_{yy} &\equiv \frac{\partial^2\tilde{U}}{\partial y^2} = 1 - \frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3} + 3(1-\mu)\frac{y^2}{\rho_1^5} + 3\mu\frac{y^2}{\rho_2^5},\\
\tilde{U}_{zz} &\equiv \frac{\partial^2\tilde{U}}{\partial z^2} = -\frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3} + 3(1-\mu)\frac{z^2}{\rho_1^5} + 3\mu\frac{z^2}{\rho_2^5},\\
\tilde{U}_{xy} &\equiv \frac{\partial^2\tilde{U}}{\partial x\partial y} = \tilde{U}_{yx} = 3\left[(1-\mu)\frac{x+\mu}{\rho_1^5} + \mu\frac{x-1+\mu}{\rho_2^5}\right]y,\\
\tilde{U}_{xz} &\equiv \frac{\partial^2\tilde{U}}{\partial x\partial z} = \tilde{U}_{zx} = 3\left[(1-\mu)\frac{x+\mu}{\rho_1^5} + \mu\frac{x-1+\mu}{\rho_2^5}\right]z,\\
\tilde{U}_{yz} &\equiv \frac{\partial^2\tilde{U}}{\partial y\partial z} = \tilde{U}_{zy} = 3\left(\frac{1-\mu}{\rho_1^5} + \frac{\mu}{\rho_2^5}\right)yz.
\end{aligned} \tag{B.1}$$

For the mixed derivatives, $\tilde{U}_{xz}$ and $\tilde{U}_{yz}$ vanish at all classical libration points because $z_{\mathcal{L}_i} = 0$. At the collinear points, $\tilde{U}_{xy}$ also vanishes because $y_{\mathcal{L}_i} = 0$, while at the triangular points $\tilde{U}_{xy}$ is generally nonzero. Moreover, at any equilibrium point, we have

$$z_{\mathcal{L}_i} = \dot{x}_{\mathcal{L}_i} = \dot{y}_{\mathcal{L}_i} = \dot{z}_{\mathcal{L}_i} = \frac{\partial\tilde{U}}{\partial x}(\mathcal{L}_i) = \frac{\partial\tilde{U}}{\partial y}(\mathcal{L}_i) = \frac{\partial\tilde{U}}{\partial z}(\mathcal{L}_i) = 0,$$

and so the linearised system reduces to

$$\begin{aligned}
\delta\ddot{x} - 2\delta\dot{y} &= \delta x\tilde{U}_{xx}(\mathcal{L}_i) + \delta y\tilde{U}_{xy}(\mathcal{L}_i),\\
\delta\ddot{y} + 2\delta\dot{x} &= \delta x\tilde{U}_{xy}(\mathcal{L}_i) + \delta y\tilde{U}_{yy}(\mathcal{L}_i),\\
\delta\ddot{z} &= \delta z\tilde{U}_{zz}(\mathcal{L}_i).
\end{aligned} \tag{B.2}$$

The general solution is a linear combination of exponential terms of the form $e^{\lambda t}$, where λ represents the eigenvalue of each term. If any eigenvalue is real, the equilibrium is *unstable*. If all eigenvalues are purely imaginary, the equilibrium is *Lyapunov* (or *centre*) *stable*, and the motion remains bounded near the equilibrium point. Since the unperturbed CR3BP is a conservative closed system, devoid of any energy dissipation mechanisms, as we will see there are no *asymptotically stable* equilibria. In addition, the term $\tilde{U}_{zz}(\mathcal{L}_i)$ is always independent of the four planar coordinates at all libration points, and thus the third equation of motion is always decoupled from the first two (as we will show in more detail below).

We can therefore interpret the eigenvalues in terms of *modal subspaces*. From the six first-order dimensions of the perturbed CR3BP,

$$(\delta x, \delta y, \delta z, \delta \dot{x}, \delta \dot{y}, \delta \dot{z}),$$

we identify two modal subspaces, namely the *vertical* $(\delta z, \delta \dot{z})$ and the *planar* $(\delta x, \delta y, \delta \dot{x}, \delta \dot{y})$. Each of the stability modes we will discuss below will therefore be associated with one of these subspaces, and the final stability descriptor will contain two planar and one vertical mode.

Collinear Points

At the collinear libration points $\mathcal{L}_1, \mathcal{L}_2$ and $\mathcal{L}_3$, the linearised equations become

$$\begin{aligned} \delta\ddot{x} - 2\delta\dot{y} &= \delta x \tilde{U}_{xx}(\mathcal{L}_i), \\ \delta\ddot{y} + 2\delta\dot{x} &= \delta y \tilde{U}_{yy}(\mathcal{L}_i), \\ \delta\ddot{z} &= \delta z \tilde{U}_{zz}(\mathcal{L}_i), \end{aligned} \tag{B.3}$$

with

$$\begin{aligned} \tilde{U}_{xx}(\mathcal{L}_i) &= 1 + \frac{2(1-\mu)}{\rho_1^3} + \frac{2\mu}{\rho_2^3}, \\ \tilde{U}_{yy}(\mathcal{L}_i) &= 1 - \frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3}, \\ \tilde{U}_{zz}(\mathcal{L}_i) &= -\frac{1-\mu}{\rho_1^3} - \frac{\mu}{\rho_2^3}, \end{aligned} \tag{B.4}$$

because $y_{\mathcal{L}_i} = z_{\mathcal{L}_i} = 0$, so $\rho_1 = |x_{\mathcal{L}_i} + \mu|$ and $\rho_2 = |x_{\mathcal{L}_i} - 1 + \mu|$ at these points.

The z–equation is decoupled from the other two (as previously anticipated), and, since the two distance terms ρ_1 and ρ_2 (from primary and secondary, respectively) are strictly positive at all collinear equilibrium points, the $\tilde{U}_{zz}(\mathcal{L}_i)$ term will always be strictly negative from inspection. Hence the motion in the vertical modal subspace is simple–harmonic (i.e. centre stable). For the remaining two components, we seek solutions of the form

$$\delta x = A_1 e^{\lambda t}, \qquad \delta y = A_2 e^{\lambda t}.$$

Substitution into the planar system Eq. (B.3) yields

$$\begin{aligned} \left[\lambda^2 - \tilde{U}_{xx}(\mathcal{L}_i)\right] A_1 - 2\lambda A_2 &= 0, \\ 2\lambda A_1 + \left[\lambda^2 - \tilde{U}_{yy}(\mathcal{L}_i)\right] A_2 &= 0, \end{aligned}$$

leading to the characteristic equation

$$\lambda^4 + 2A_3\lambda^2 - A_4 4 = 0, \tag{B.5}$$

where

$$A_3 = 2 - \frac{1}{2}\Big[\tilde{U}_{xx}(\mathcal{L}_i) + \tilde{U}_{yy}(\mathcal{L}_i)\Big],$$
$$A_4 = -\tilde{U}_{xx}(\mathcal{L}_i)\tilde{U}_{yy}(\mathcal{L}_i).$$

The roots of Eq. (B.5) are given by

$$\lambda_{1,2,3,4} = \pm\sqrt{-A_3 \pm \sqrt{A_3^2 + A_4}},$$

To determine the nature of these eigenvalues, note that the sign of term $-A_3 \pm \sqrt{A_3^2 + A_4}$ is dictated by the sign of A_4. Define

$$a \coloneqq \frac{1-\mu}{\rho_1^3} + \frac{\mu}{\rho_2^3}.$$

Then

$$\tilde{U}_{xx}(\mathcal{L}_i) = 1 + 2a, \qquad \tilde{U}_{yy}(\mathcal{L}_i) = 1 - a, \qquad \tilde{U}_{zz}(\mathcal{L}_i) = -a. \tag{B.6}$$

We have $a > 0$ since $\rho_1, \rho_2 > 0$ at all libration points. From inspection, term $\tilde{U}_{xx}(\mathcal{L}_i)$ is then always strictly positive. Now consider term $\tilde{U}_{yy}(\mathcal{L}_i)$. To determine the sign of this term, it is sufficient to check whether a is greater than or smaller than 1 at each collinear point. For $\mathcal{L}_1$, the point lies between the primary and secondary, so both distances satisfy

$$\rho_1 < 1, \qquad \rho_2 < 1.$$

Therefore,

$$\frac{1-\mu}{\rho_1^3} > 1 - \mu, \qquad \frac{\mu}{\rho_2^3} > \mu,$$

and hence

$$a = \frac{1-\mu}{\rho_1^3} + \frac{\mu}{\rho_2^3} > (1-\mu) + \mu = 1.$$

For $\mathcal{L}_2$, write the distance from $\mathcal{L}_2$ to the primary as $\gamma > 0$. Then

$$\rho_2 = \gamma, \qquad \rho_1 = 1 + \gamma,$$

and the collinear equilibrium condition Eq. (3.4) gives

$$1 - \mu + \gamma - \frac{1-\mu}{(1+\gamma)^2} - \frac{\mu}{\gamma^2} = 0.$$

Rearranging and substituting into $a - 1$ gives

$$\begin{aligned} a - 1 &= \frac{1-\mu}{(1+\gamma)^3} + \frac{\mu}{\gamma^3} - 1 \\ &= \frac{1-\mu}{(1+\gamma)^3} + \left[\frac{1-\mu}{\gamma} + 1 - \frac{1-\mu}{\gamma(1+\gamma)^2}\right] - 1 \\ &= (1-\mu)\left[\frac{1}{(1+\gamma)^3} + \frac{1}{\gamma} - \frac{1}{\gamma(1+\gamma)^2}\right]. \end{aligned}$$

The term in brackets is positive for $\gamma > 0$, so $a > 1$ at $\mathcal{L}_2$.

For $\mathcal{L}_3$, write the distance from $\mathcal{L}_3$ to the primary as $\gamma > 0$. Then

$$\rho_1 = \gamma, \qquad \rho_2 = 1 + \gamma,$$

and the collinear equilibrium condition Eq. (3.4) gives

$$-\mu - \gamma + \frac{1-\mu}{\gamma^2} + \frac{\mu}{(1+\gamma)^2} = 0. \tag{B.7}$$

Rearranging and substituting into $a - 1$ gives

$$\begin{aligned} a - 1 &= \frac{1-\mu}{\gamma^3} + \frac{\mu}{(1+\gamma)^3} - 1 \\ &= \left[\frac{\mu}{\gamma} + 1 - \frac{\mu}{\gamma(1+\gamma)^2}\right] + \frac{\mu}{(1+\gamma)^3} - 1 \\ &= \mu\left[\frac{1}{\gamma} - \frac{1}{\gamma(1+\gamma)^2} + \frac{1}{(1+\gamma)^3}\right]. \end{aligned}$$

Again, the term in brackets is positive for $\gamma > 0$, so $a > 1$ at $\mathcal{L}_3$. Therefore, for all three collinear libration points, $a > 1$. It follows that

$$\tilde{U}_{yy}(\mathcal{L}_i) = 1 - a < 0, \tag{B.8}$$

and consequently,

$$A_4 = -\tilde{U}_{xx}(\mathcal{L}_i)\tilde{U}_{yy}(\mathcal{L}_i) > 0. \tag{B.9}$$

Thus the term $-A_3 \pm \sqrt{A_3^2 + A_4}$ will always yield two oppositely signed values (since $|\sqrt{A_3^2 + A_4}| > |A_3|$). As a result, the roots $\lambda_{1,2,3,4}$ of Eq. (B.5) will always consist in two real and two purely imaginary values.

Combined with the harmonic motion along the vertical modal subspace, the full spatial linearised dynamics at the collinear points consist of one real eigenvalue pair and two purely imaginary eigenvalue pairs. Hence the collinear libration points have a planar saddle, a planar centre, and a vertical centre, written in short as

$$\textit{centre} \times \textit{centre} \times \textit{saddle}.$$

Triangular Points

For the triangular libration points $\mathcal{L}_4$ and $\mathcal{L}_5$, the linearised system is

$$\begin{aligned} \delta\ddot{x} - 2\delta\dot{y} &= \frac{3}{4}\delta x + \frac{\sqrt{27}}{4}(1-2\mu)\delta y, \\ \delta\ddot{y} + 2\delta\dot{x} &= \frac{\sqrt{27}}{4}(1-2\mu)\delta x + \frac{9}{4}\delta y, \\ \delta\ddot{z} &= -\delta z. \end{aligned}$$

In the z–direction the motion is again harmonic, so the vertical modal subspace is stable. For the planar part, the characteristic equation becomes

$$\lambda^4 + \lambda^2 + \frac{27}{4}\mu(1-\mu) = 0. \tag{B.10}$$

The roots of Eq. (B.10) are given by

$$\lambda_{1,2,3,4} = \pm\sqrt{\frac{-1 \pm \sqrt{1 - 27\mu(1-\mu)}}{2}}.$$

If the condition

$$1 - 27\mu(1-\mu) > 0$$

is met, then both values of λ^2 are real and negative, so the planar eigenvalues are purely imaginary. The threshold value for this inequality is known as the *Routh critical mass parameter* μ_R, satisfying

$$1 - 27\mu_R(1-\mu_R) = 0.$$

Since by definition we have $m_2 \leq m_1$, the mass ratio is constrained by $0 < \mu \leq 1/2$. This gives

$$\mu_R = \frac{1}{2}\left(1 - \sqrt{\frac{23}{27}}\right) = \frac{1}{2} - \frac{\sqrt{69}}{18} \approx 0.0385. \tag{B.11}$$

For all eight[3] of the Solar System's planets, we have $\mu < \mu_R$ in the Sun–planet system (in fact, the largest of the mass ratios is that of Jupiter at 9.5388×10^{-4}), so the triangular libration points are always Lyapunov stable. This is also true of the Earth–Moon system. Combined with the harmonic motion along the vertical modal subspace, the full linearised dynamics at the triangular points consist of three purely imaginary eigenvalue pairs. Hence, for $\mu < \mu_R$, the triangular libration points have two planar centres and one vertical centre, written in short as

$$\textit{centre} \times \textit{centre} \times \textit{centre}.$$

At $\mu = \mu_R$, the two planar imaginary eigenvalue pairs coalesce on the imaginary axis, producing a degenerate 1:1 resonance. This marks a Hamiltonian–Hopf bifurcation, also called a Krein collision, at which the triangular points lose linear stability. For $\mu > \mu_R$, the planar eigenvalues leave the imaginary axis as a complex quartet with non-zero real parts, so the triangular libration points become linearly unstable.

Remark

Among the eight planets and their satellites, the Earth–Moon system has by far the largest mass ratio (defined in Table 1). Thus, no planet–satellite system in the Solar System comes close to violating the Routh mass threshold. However, things are different in the trans-Neptunian domain. Because planetesimal collisions in the early outer solar system frequently resulted in nearly equal-mass binary pairs, there are actually a few examples of "planet"–satellite systems where the triangular points $\mathcal{L}_4$ and $\mathcal{L}_5$ are unstable. Among them is Pluto with its satellite, Charon! The Pluto–Charon system has, in fact, a mass ratio of $\mu \approx 0.108$, which exceeds the Routh mass threshold. Given the near-zero eccentricity of Charon's orbit around Pluto, and the small external perturbations due to the limited size of Pluto's other satellites and Pluto's enormous distance from the Sun, the CR3BP provides a very accurate model for this system. In fact, when NASA was planning the New Horizons flyby of Pluto in 2015, astrodynamicists relied heavily on CR3BP models [Ste08]. Because the triangular $\mathcal{L}_4$ and $\mathcal{L}_5$ points are violently unstable, engineers had to model the chaotic regions to ensure the spacecraft wouldn't accidentally smash into a hidden ring of debris or a stray moonlet that had been ejected from a stable orbit into a chaotic manifold.

[3]Sorry, Pluto.

C The Crash Region

For certain initial conditions, an orbiting body may quickly enter Earth's atmosphere or crash into the lunar surface. From a practical perspective, points in the state space leading to these outcomes are not very interesting (or at least, need little consideration), as an object will quickly de-orbit when placed under initial conditions corresponding to them.

We construct a set consisting of all points that will lead to such results, and refer to it as the *crash region*.

Indeed, in the two-dimensionally projected case, this region is analytically derived in [Grä25], using techniques from [AM87], as the area bounded by the curve

$$(v_y + x)^2 = \frac{R_i^2 v_x^2 + 2m_i R_i \left(1 - \frac{R_i}{|x-\rho_i|}\right)}{|x-\rho_i|^2 - R_i^2}, \tag{C.1}$$

where the normalised masses of the Earth and Moon are denoted by m_1 and m_2 as defined in Section A, $R_1 \approx 7000\,\text{km/DU}$ and $R_2 = 1738.1\,\text{km/DU}$ denote the radii of the Earth (including a significant portion of the atmosphere) and the Moon, and ρ_1, ρ_2 are the distances of the Earth/Moon from the system barycentre, respectively. The Earth and Moon crash regions can both be seen in Fig. 11. Previous work in [Grä25] has numerically verified the validity of this analytically-derived region by testing which initial conditions lead to crashes within a single orbital period.

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